

Retail Analytics Report

Revenue Forecasting, Customer Churn Prediction & Customer Lifetime Value Analysis

UCI Online Retail II — UK E-Commerce Dataset

Analysis period: December 2009 – December 2010

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Section 1 — Executive Summary

This report presents a complete retail analytics study of a UK-based e-commerce business operating across the December 2009 to December 2010 period. The analysis spans four interconnected workstreams: exploratory data analysis, revenue forecasting, customer churn prediction, and customer lifetime value modelling. Together they answer one business question: **where is revenue at risk, which customers represent it, and what should be done about it.**

Key findings at a glance

Metric	Value
Total revenue analysed	£10,271,762
Unique registered customers	4,383
Repeat customer rate	67.09%
Top 20% of customers	73.5% of revenue
Overall churn rate	35.9%
Revenue at risk (churn model)	£108,351 per period
Expected revenue saved at 35% retention	£46,308 per period
CLV model accuracy (R^2)	0.684
Revenue forecast — conservative	£840,410
Revenue forecast — growth scenario	£1,464,293

Three things that can be acted on immediately, without waiting for any model

1. Customers inactive for 60 days churn at 53.7%. A reactivation trigger at day 45 requires no machine learning to implement.
2. New customers churn at 43.9% — nearly double the average. Onboarding is broken and fixable.
3. Customers who return items are more loyal, not less. Any policy suppressing communication to high-returners should be reviewed.

The churn prediction and CLV models then identify exactly which customers these rules apply to, allowing the business to prioritise where human attention and budget go.

Section 2 — Dataset Overview

The dataset is sourced from the UCI Online Retail II repository and covers transactions from a UK-based wholesale gift and homeware retailer. The raw dataset contained 525,461 transaction rows across eight fields: invoice number, stock code, product description, quantity, invoice date, unit price, customer ID, and country.

Data quality summary

Issue	Count	Action taken
Duplicate rows	6,865	Removed
Missing descriptions	2,928 (0.56%)	Removed
Missing customer IDs	107,927 (20.5%)	Flagged as guest transactions
Cancellation invoices (prefix C)	10,182	Flagged, excluded from valid sales
Zero or negative price rows	756	Excluded from merchandise analysis
Non-merchandise SKUs (postage, fees)	12 SKU codes	Excluded from product analysis

After cleaning, the working dataset contained **515,668 rows** with **504,730 valid sales transactions**.

Dataset scope

- Date range: 1 December 2009 to 9 December 2010
- Unique invoices: 25,888
- Unique products: 4,276
- Unique registered customers: 4,383
- Countries covered: 40 (United Kingdom dominant)

Note on guest transactions: 20.5% of rows carry no customer ID. These are treated as guest purchases throughout the analysis and excluded from customer-level modelling (churn, CLV). They are included in revenue and product-level calculations.

Section 3 — Exploratory Data Analysis

3.1 Headline KPIs

Metric	All Transactions	Merchandise Only
Total revenue	£10,271,762	£9,806,616
Total orders	20,951	20,585
Average order value	£490.28	£476.40
Median order value	—	£298.65
Repeat customer rate	67.09%	—

The gap between mean AOV (£476) and median AOV (£299) indicates a right-skewed order distribution — a small number of large bulk orders pull the average up significantly. The median is the more representative figure for typical customer behaviour.

Guest vs Registered customers

Segment	Revenue	Orders	AOV
Registered	£8,640,613	18,978	£455.30
Guest	£1,166,003	1,607	£725.58

Guest customers show a dramatically higher AOV (£725 vs £455). This likely reflects the B2B nature of guest purchases — large one-off wholesale orders placed without account registration.

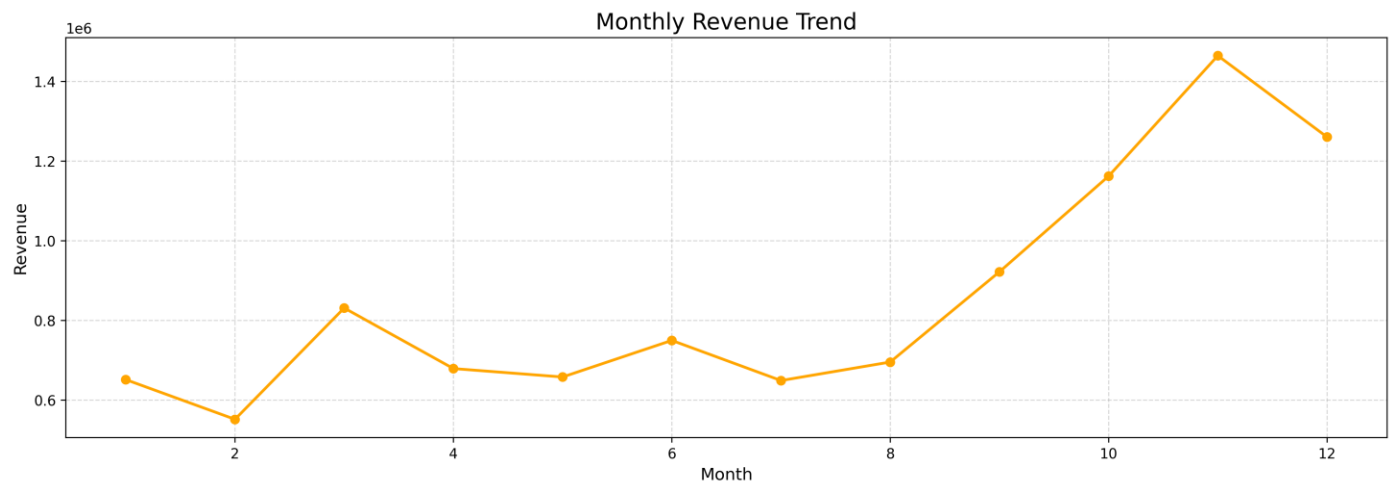


Figure 1 — Monthly Revenue Trend

The monthly revenue chart reveals a business with moderate baseline stability between £550K and £950K for most of the year, followed by a sharp spike in November 2010 (£1,464,293) and an equally

sharp collapse in December 2010 (£437,631). The November spike is 99.5% above the long-run average. The December figure reflects a partial month — the dataset ends 9 December — rather than genuine demand collapse.

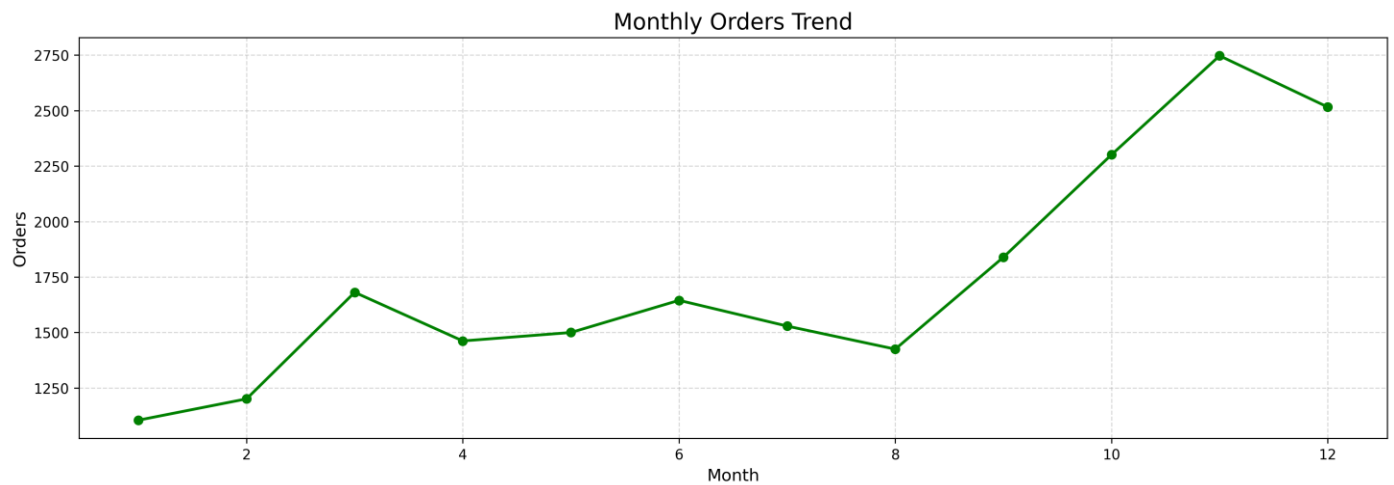


Figure 2 — Monthly Orders Trend

3.2 Revenue by Day and Hour

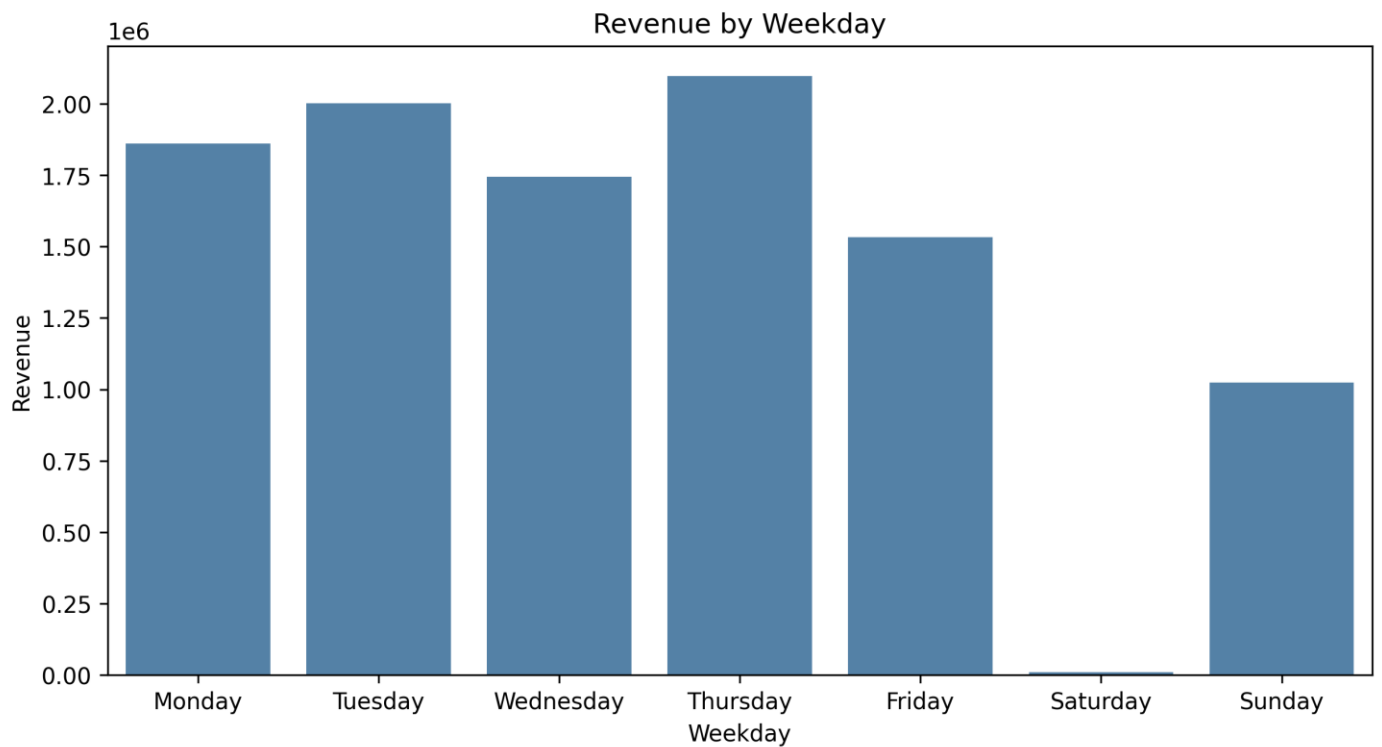


Figure 3 — Revenue by Weekday

Revenue is heavily concentrated on weekdays, consistent with a B2B wholesale customer base placing orders during business hours. Weekend revenue is negligible. Thursday generates the highest weekday revenue.

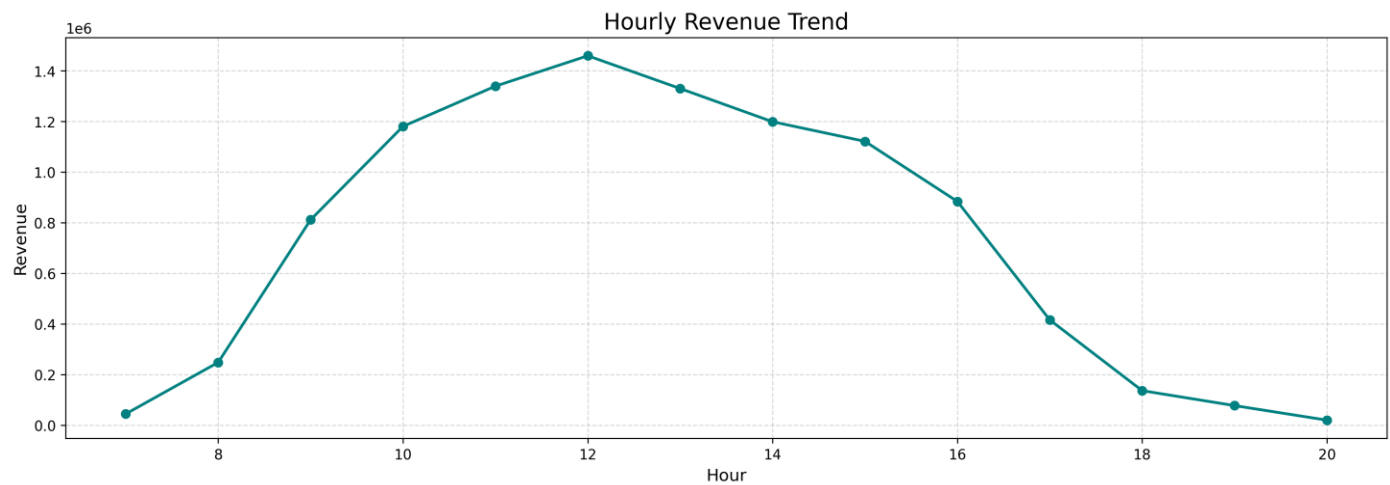


Figure 4 — Hourly Revenue Trend

The peak revenue window is 10:00–14:00, with the single highest revenue hour at 12:00. Orders placed outside business hours (before 8:00 or after 17:00) account for a minimal share of total revenue.

3.3 Customer Segmentation — RFM Analysis

Customers were segmented using RFM scoring (Recency, Frequency, Monetary) across five quartiles. Segments were then collapsed into five business-relevant groups based on score combinations.

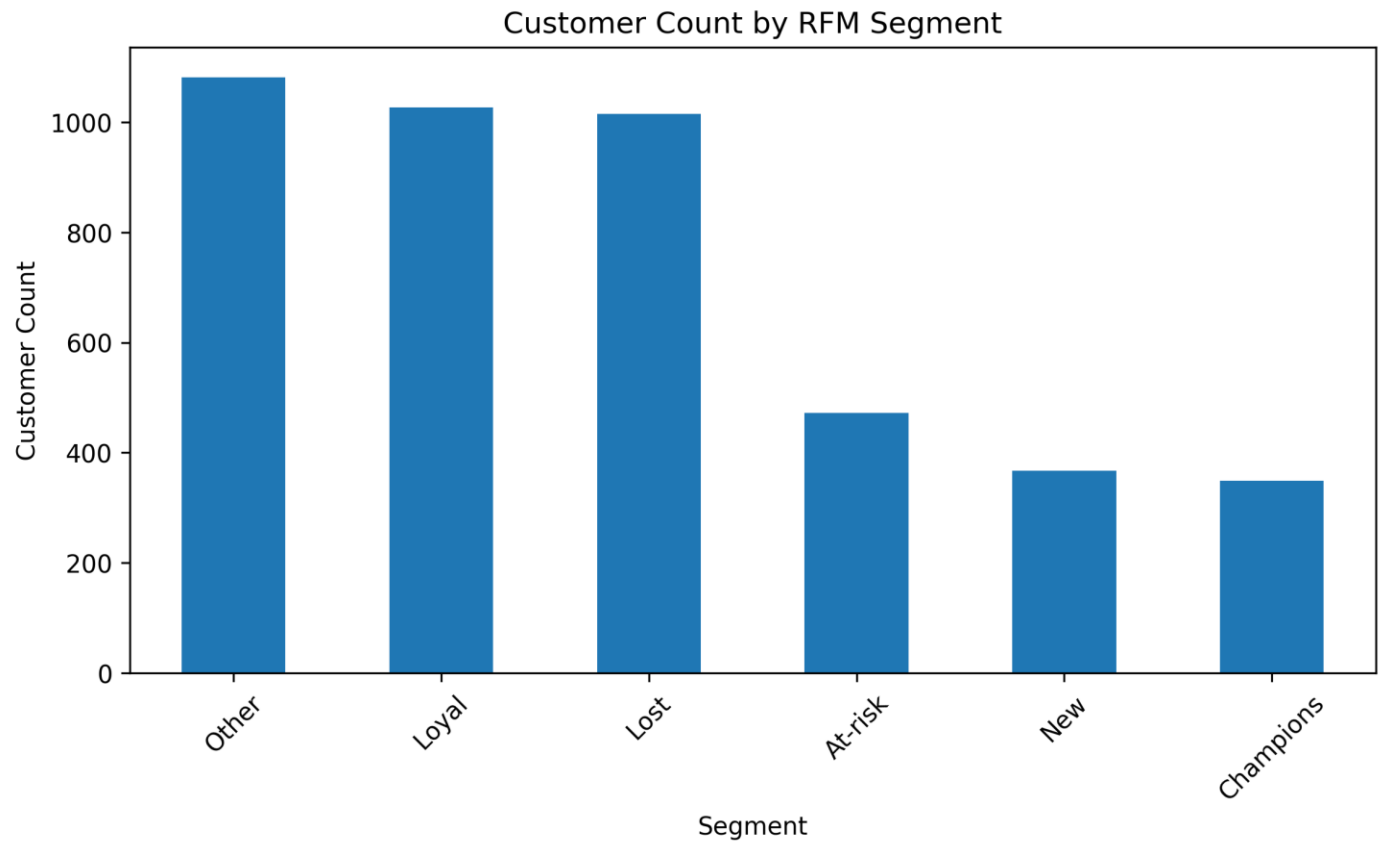


Figure 5 — Customer Count by RFM Segment

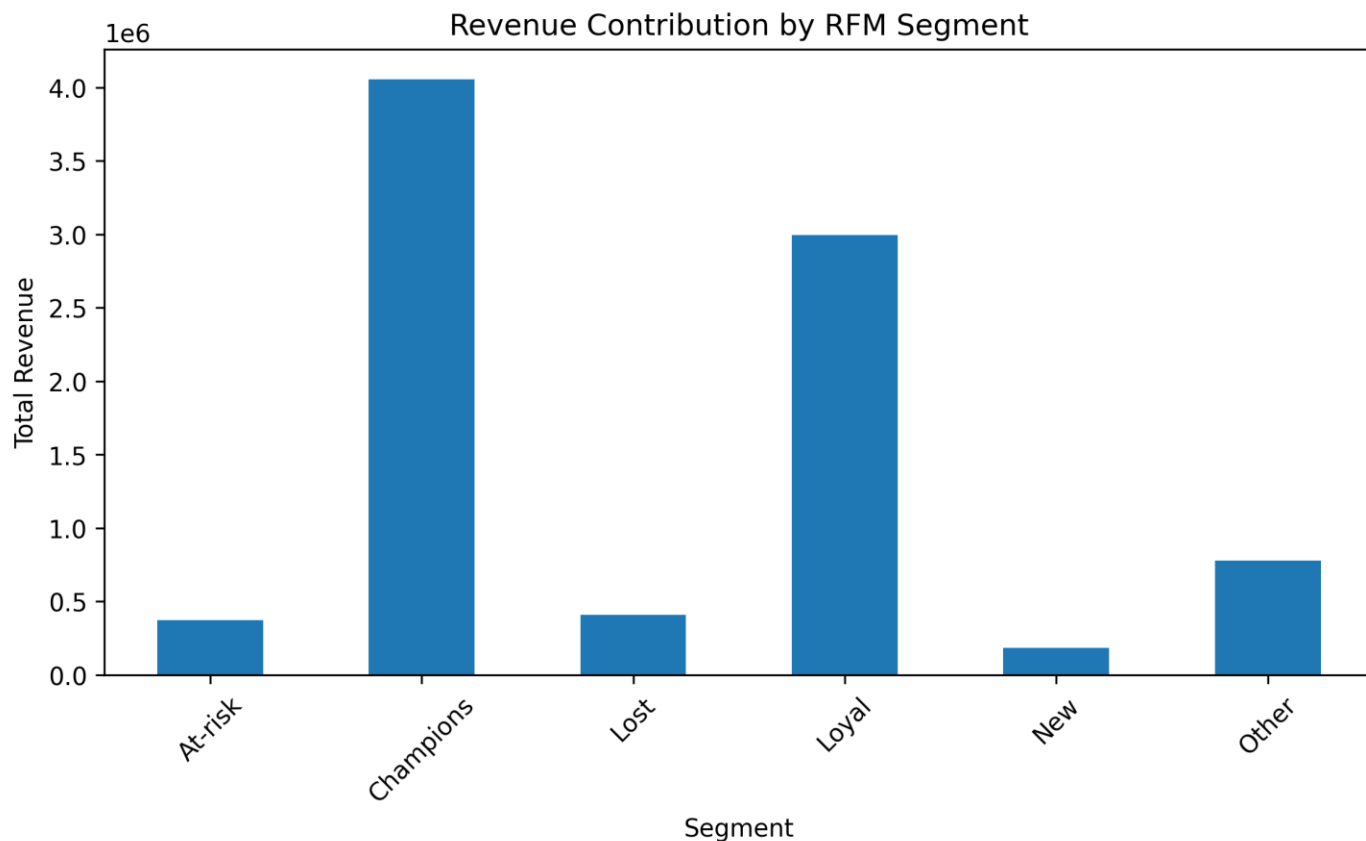


Figure 6 — Revenue Contribution by RFM Segment

Pareto concentration: the top 20% of customers contribute 73.5% of total revenue. This has a direct implication for all modelling work — a model that catches the average churner but misses the high-value segment is commercially dangerous regardless of its aggregate accuracy metrics.

3.4 Guest vs Registered Revenue Trend

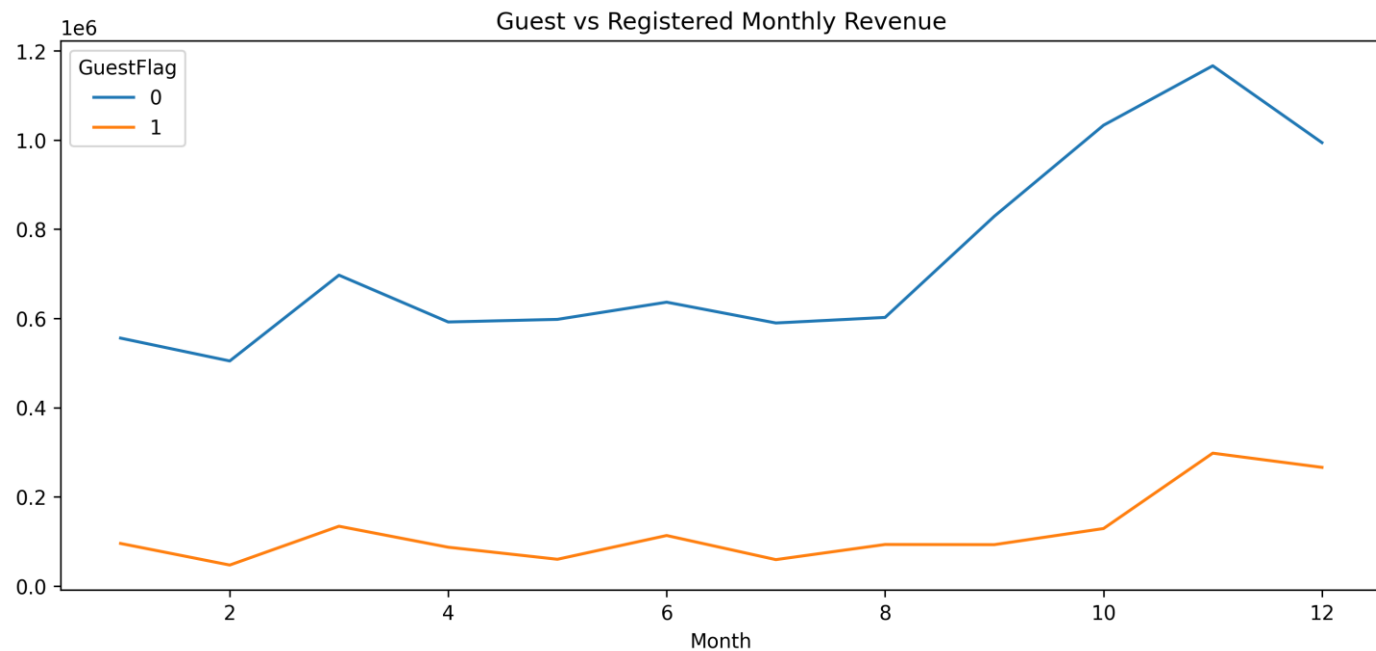


Figure 7 — Guest vs Registered Monthly Revenue

Registered customer revenue dominates in every month. Guest revenue shows spikes in certain months consistent with large one-off orders rather than recurring purchasing behaviour.

Section 4 — Revenue Diagnostics

4.1 Monthly Revenue Decomposition

To understand why revenue changes month-to-month, each monthly movement was decomposed into two components: the contribution from order volume changes and the contribution from average order value (AOV) changes.

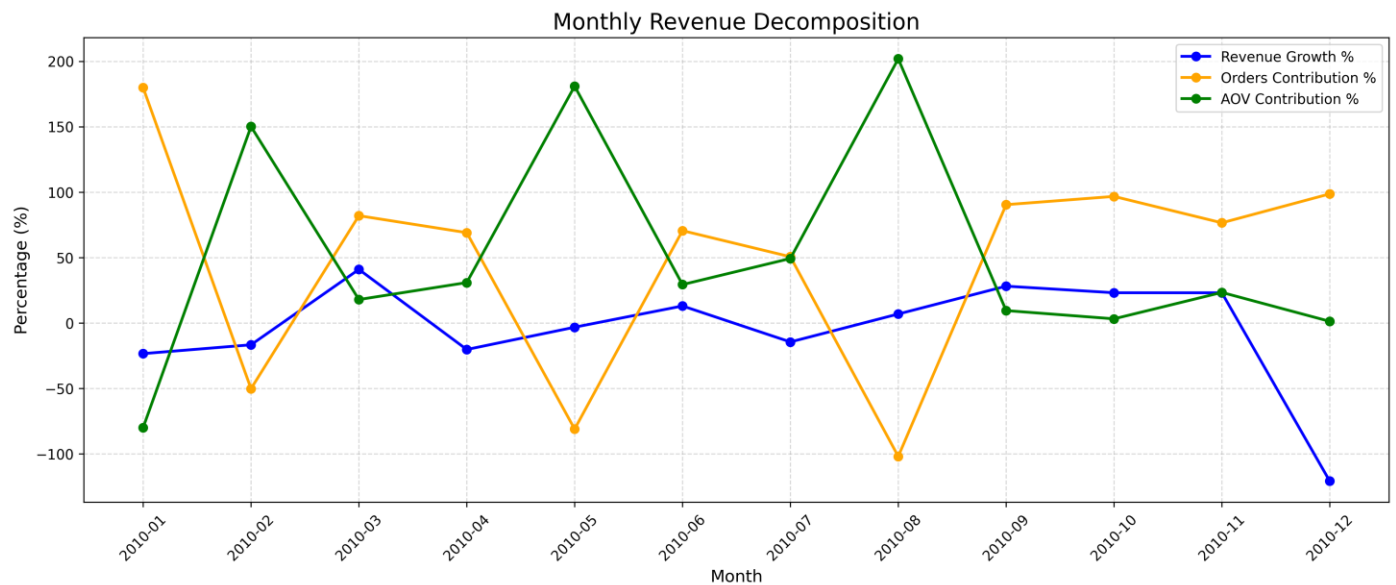


Figure 8 — Monthly Revenue Decomposition

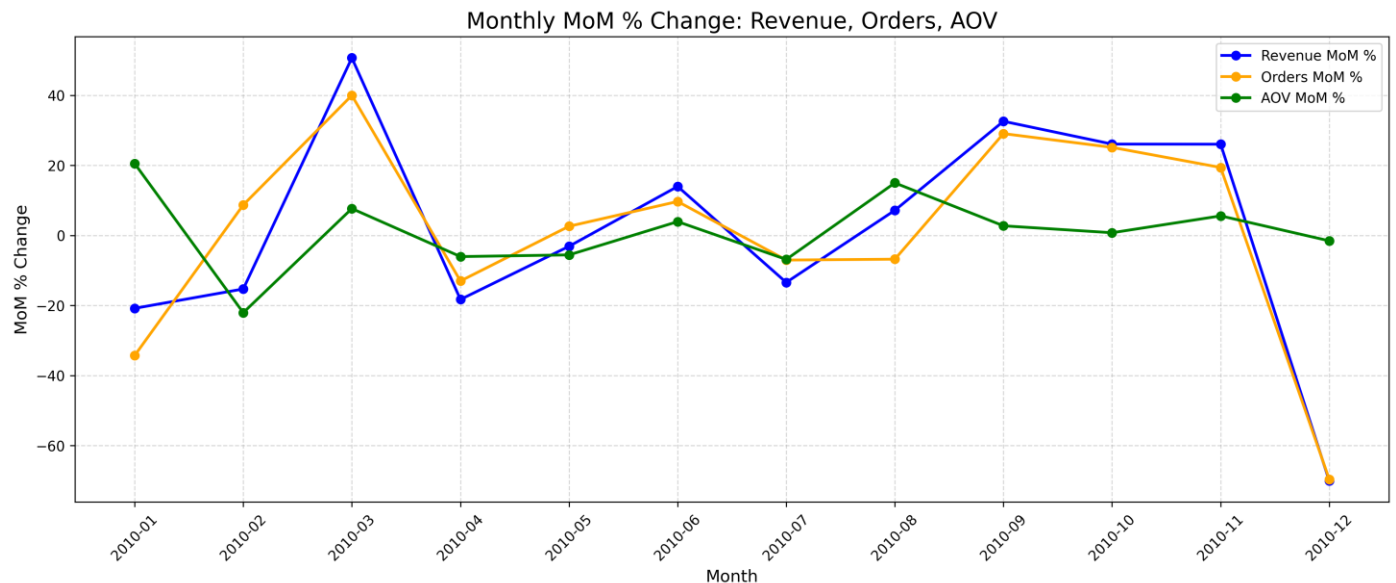


Figure 9 — Month-over-Month % Change

Key findings from decomposition

Over the 12-month period, revenue exhibits moderate stability but fluctuations are primarily driven by **Average Order Value rather than order volume**. Months such as January, February, and August show pronounced AOV spikes, suggesting impacts from bulk purchasing, promotions, or seasonal effects rather than genuine changes in customer demand.

The December revenue decline (-120.78% decomposition figure) is explained by the dataset ending on 9 December — only 9 days of data versus a full month, meaning orders and revenue are structurally truncated, not reflective of actual December trading conditions.

Month	Revenue Growth	Orders Contribution	AOV Contribution
Jan 2010	-23.4%	+179.9%	-79.9%
Mar 2010	+41.0%	+82.0%	+18.0%
Sep 2010	+28.2%	+90.5%	+9.5%
Oct 2010	+23.2%	+96.8%	+3.2%
Nov 2010	+23.1%	+76.6%	+23.4%

September through November shows a consistent growth pattern driven primarily by increasing order volume — the most commercially sustainable form of revenue growth.

Section 5 — Product & Returns Analysis

5.1 Top Revenue Products

The top five merchandise products by revenue:

Product	Revenue
Regency Cakestand 3 Tier	£169,912
White Hanging Heart T-Light Holder	£158,305
Assorted Colour Bird Ornament	£72,890
Paper Chain Kit 50s Christmas	£58,127
Jumbo Bag Red Retrospot	£54,368

Non-merchandise SKUs (postage, bank charges, adjustments) account for £465,146 — 4.53% of gross revenue. These are excluded from all product-level analysis.

5.2 Return Behaviour

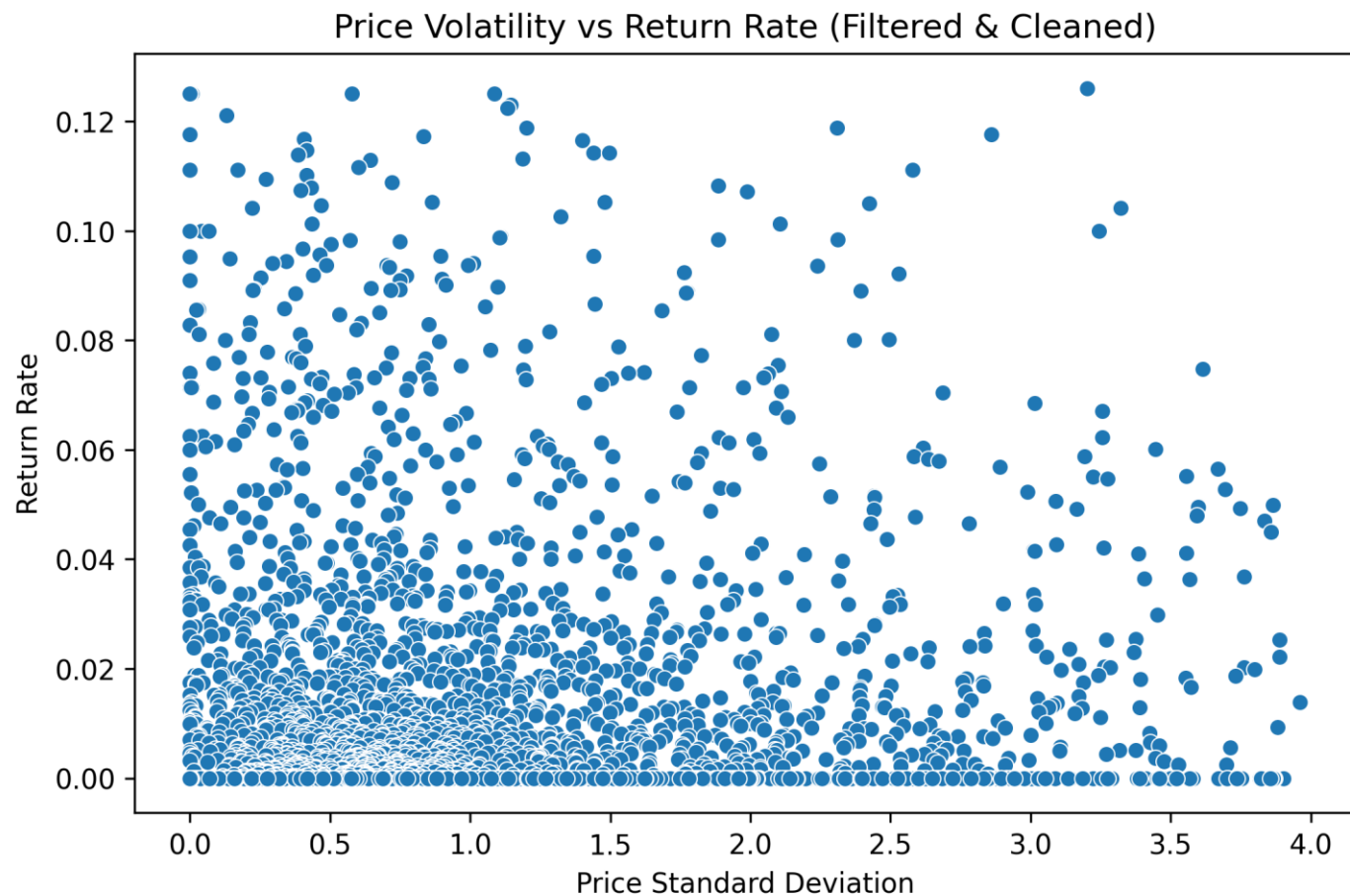


Figure 10 — Price Volatility vs Return Rate

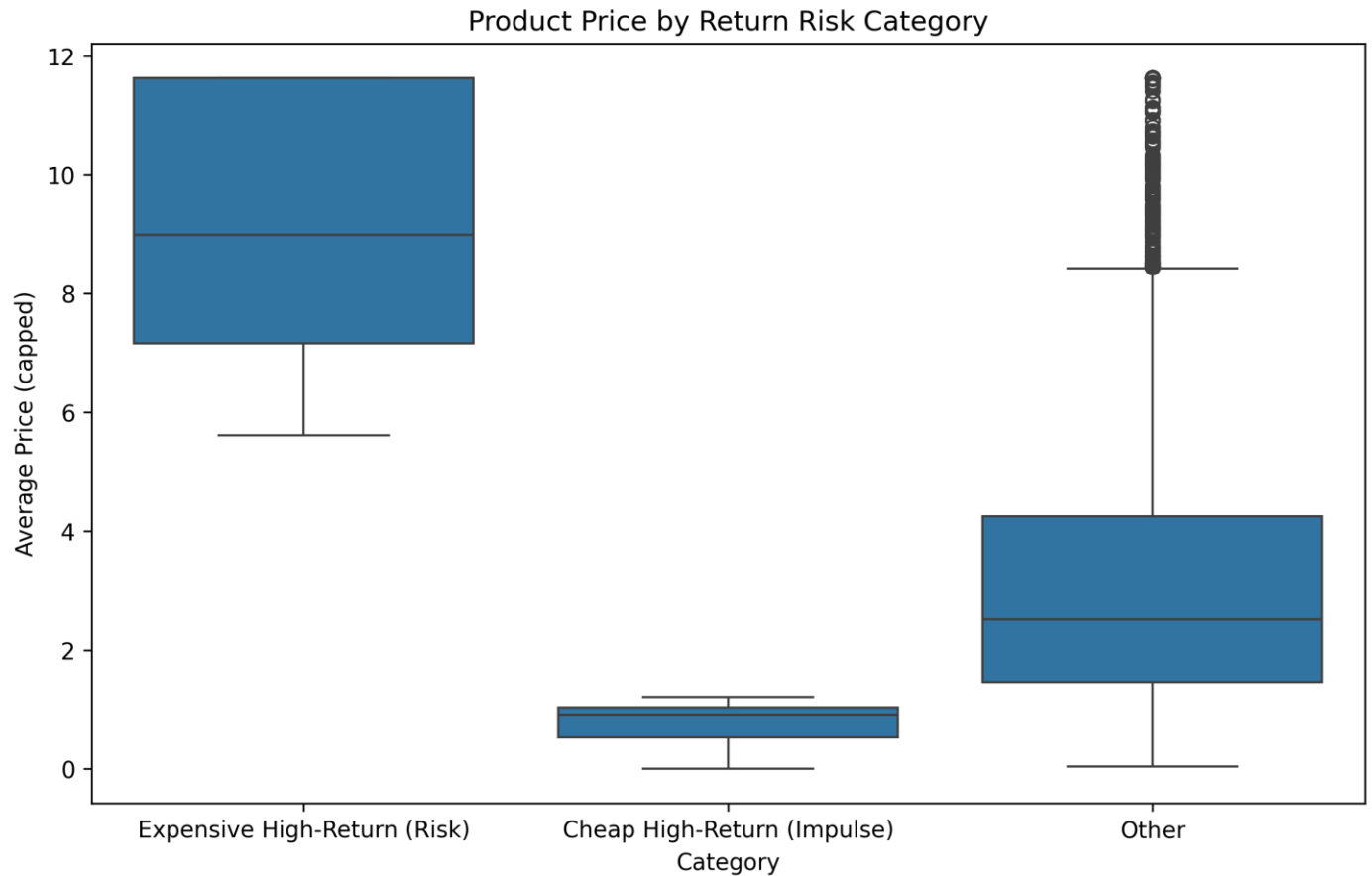


Figure 11 — Product Price by Return Risk Category

Critical finding: price does not drive returns. After controlling for sales volume and excluding non-merchandise items:

- Correlation between average price and return rate: **0.072** (negligible)
- Correlation between price volatility and return rate: **0.120** (negligible)
- High-return products average price: £4.02
- Low-return products average price: £4.05

Returns are not a pricing problem. They are driven by operational factors — product fit, fulfilment quality, or customer expectation mismatch.

5.3 Bulk vs Single-Unit Returns

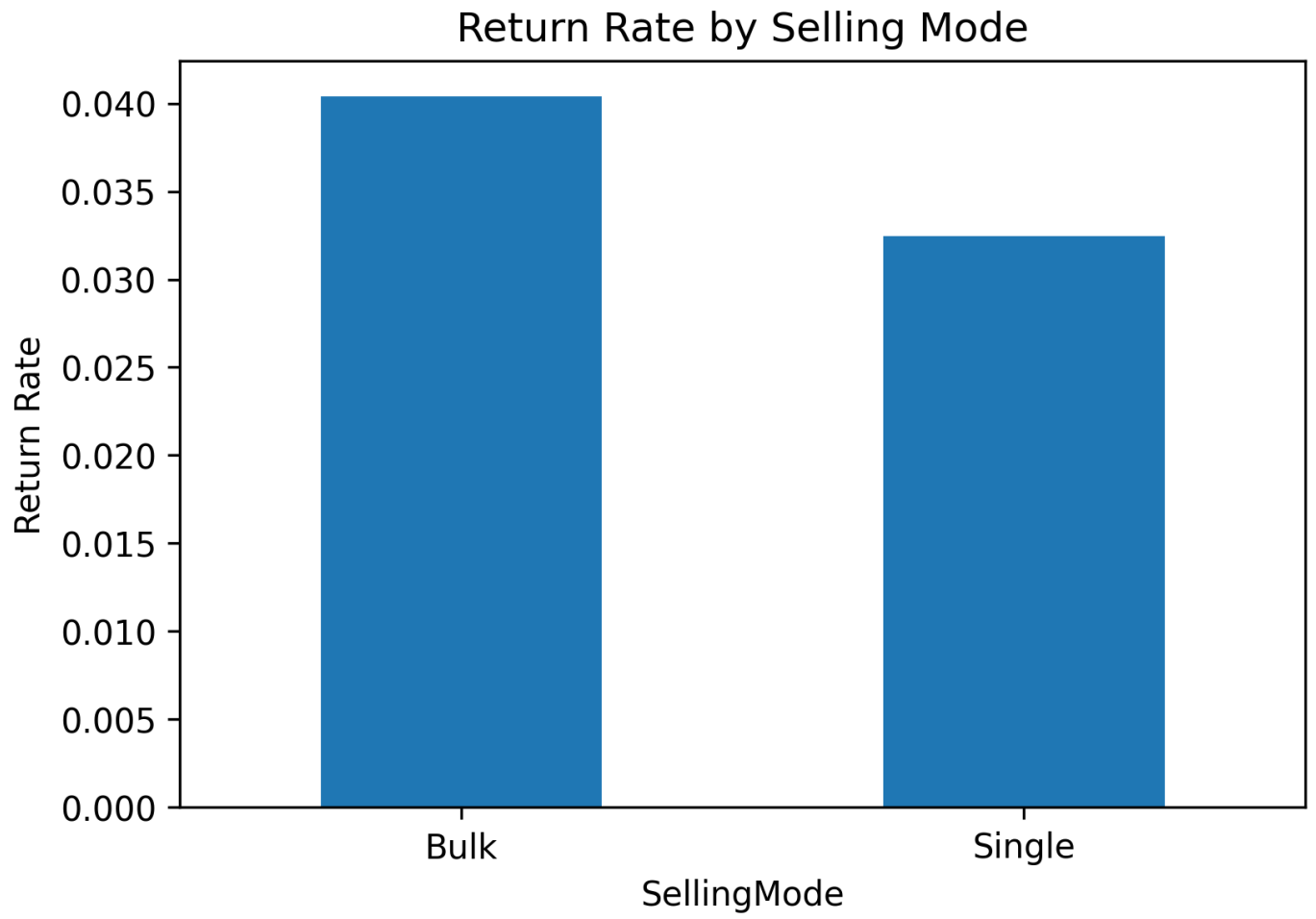


Figure 12 — Return Rate by Selling Mode

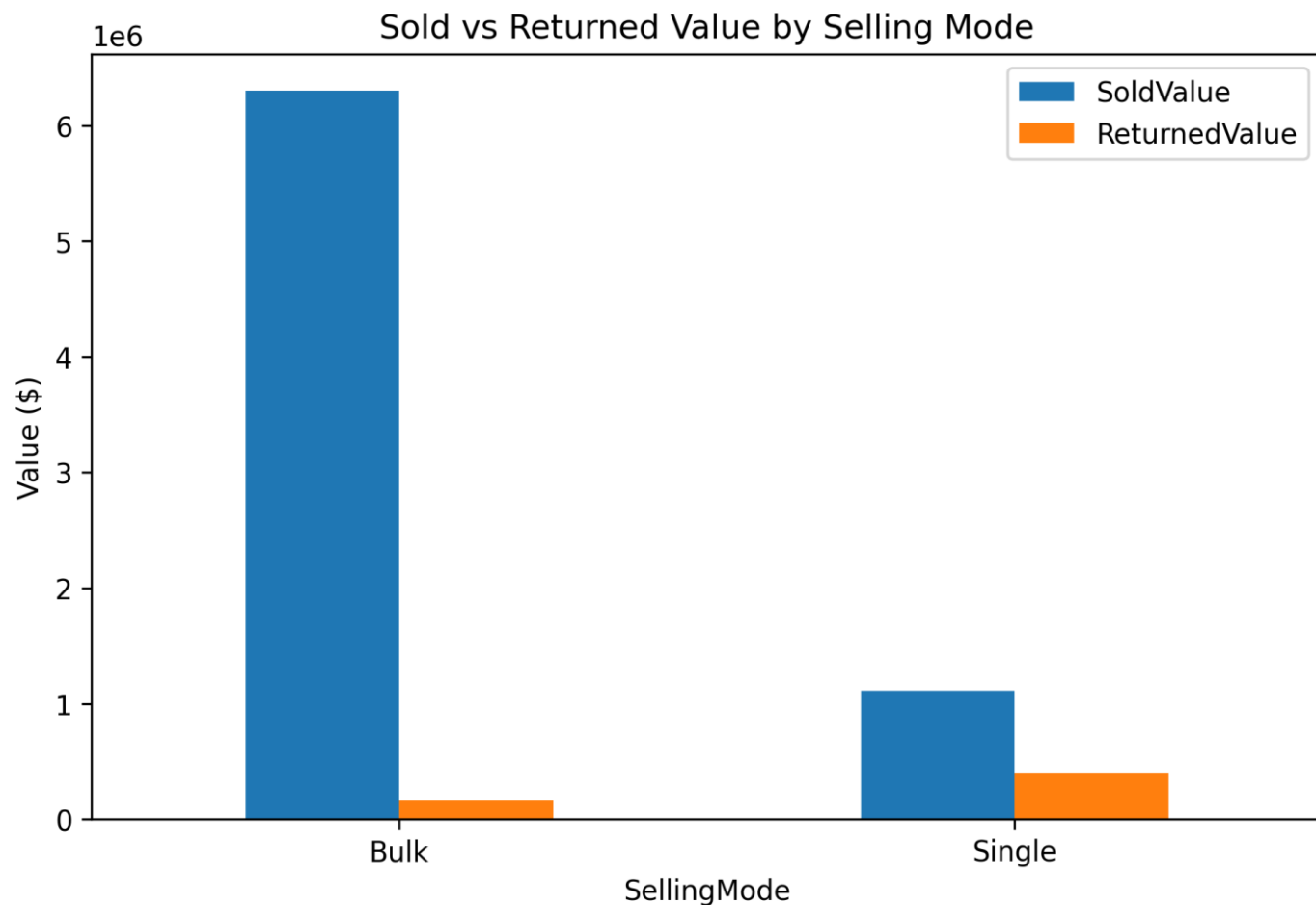


Figure 13 — Sold vs Returned Value by Selling Mode

Bulk purchases account for the vast majority of both sales volume and returned units. While bulk orders carry a slightly higher return rate than single-unit purchases, the difference is modest. The operational implication is significant: even a small improvement in bulk return rates has a disproportionate impact on revenue recovery and logistics costs.

Section 6 — Revenue Forecasting

Important caveat — read before interpreting any number in this section. This forecast is based on 13 monthly observations with only 6 usable validation points. A reliable confidence interval requires a minimum of 24 months of data. The planning range presented here is honest and wide — this reflects data scarcity, not a modelling failure. All figures are directional only and should not be used for budget commitments or financial reporting without additional data.

6.1 Historical Revenue with Structural Shocks

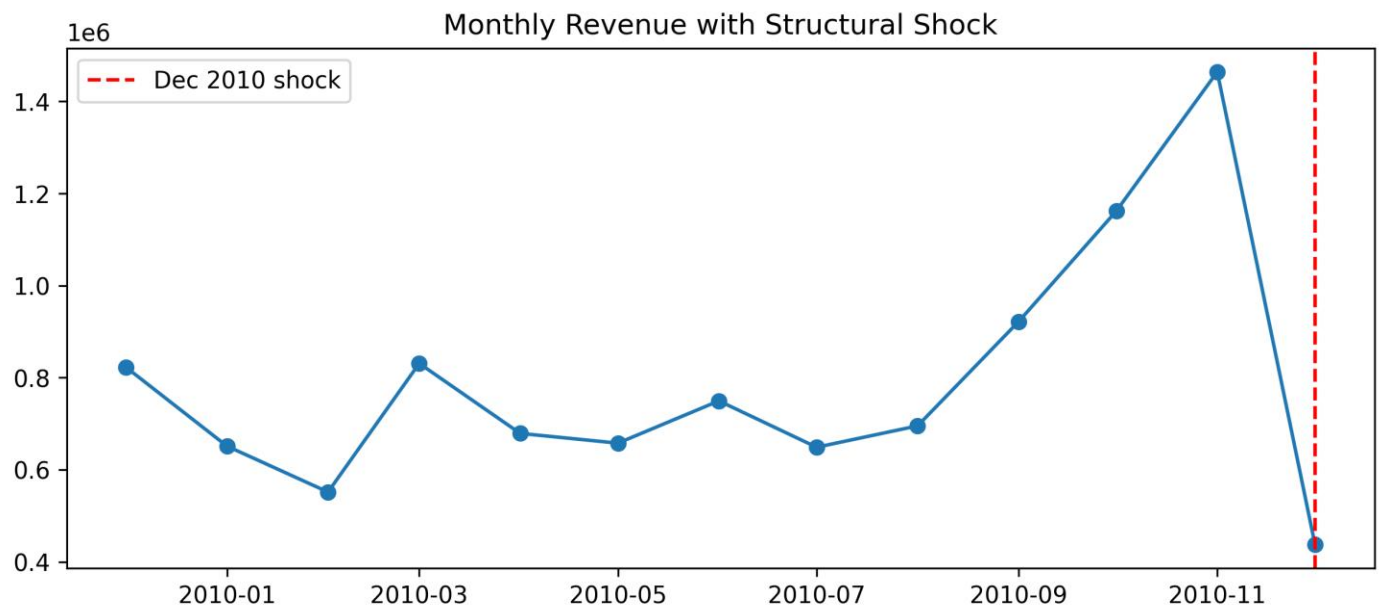


Figure 14 — Monthly Revenue with Structural Shock

Two structural anomalies are visible in the revenue series:

November 2010 — High outlier: £1,464,293 revenue, z-score of 2.60, representing a 99.5% spike above the long-run average of £733,956. The cause cannot be determined from transaction data alone. Two interpretations are possible: a large one-time order or genuine accelerating demand.

December 2010 — Low outlier: £437,631 revenue, representing a 46.6% drop below the long-run average. This is a **data artefact** — the dataset ends 9 December, meaning only 9 days of orders are captured versus a full month. This is not genuine demand collapse. Order volume in December fell 69.6% month-on-month, entirely consistent with a partial month, not a business disruption.

6.2 Dual Scenario Forecast

Because November's status (genuine growth vs temporary anomaly) cannot be determined from 13 months of data, two forecast scenarios are presented. Using a single-point forecast would misrepresent the actual uncertainty.

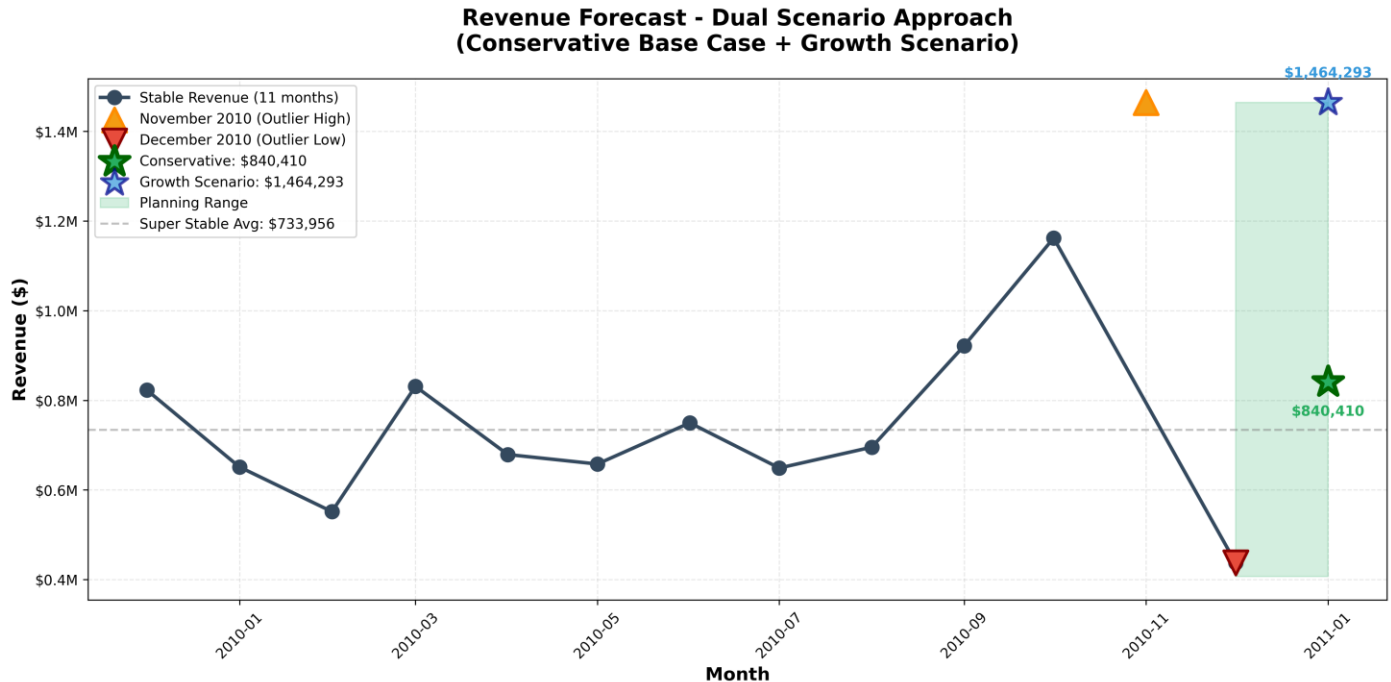


Figure 15 — Dual Scenario Forecast Chart

Scenario	Model	Forecast	Assumption
Conservative (base case)	MA(3) on stable data	£840,410	November anomalous — reversion to trend
Growth	Naive (last observation)	£1,464,293	November genuine growth continues
Worst case	Conservative + Dec shock	£458,530	December-type disruption repeats

Planning range: £407,019 — £1,464,293

The conservative scenario is recommended as the primary planning figure. It averages the three most recent stable months and avoids anchoring to an outlier. If January 2011 actual revenue exceeds £1,300,000, the growth scenario is confirmed and the forecast should be revised upward.

Revenue volatility: Coefficient of Variation = 30.3% — this is high. A business with 30% monthly revenue volatility requires flexible cost structures and cash reserves rather than fixed budget commitments.

6.3 Decision Framework

January Actual	Interpretation	February Forecast
Above £1,400,000	Growth confirmed	Switch to Naive model; forecast £1,400,000+
£1,200,000 – £1,400,000	Stable operations	Continue MA(3); forecast £1,200,000 – £1,300,000

January Actual	Interpretation	February Forecast
Below £1,200,000	Potential downturn	Investigate immediately; apply weighted MA

Section 7 — Customer Churn Prediction

7.1 Problem Definition

Churn was defined behaviourally: customers who purchased during the observation window (December 2009 — June 2010) but made no purchase in the churn assessment window (July — September 2010) were labelled as churned. This produced:

- Total customers modelled: 1,692
- Churned (1): 608 customers — 35.9%
- Retained (0): 1,084 customers — 64.1%

Label noise note: The churn model defined inactivity over the assessment window described above. The CLV analysis later revealed that only 307 customers (18.1%) had zero spend in the 6-month prediction window — meaning 301 customers labelled as "churned" subsequently returned and purchased. This is an inherent limitation of behavioural churn labelling and is expected to slightly understate true model performance. Confirmed cancellation tracking would eliminate this in future iterations.

7.2 Key EDA Findings

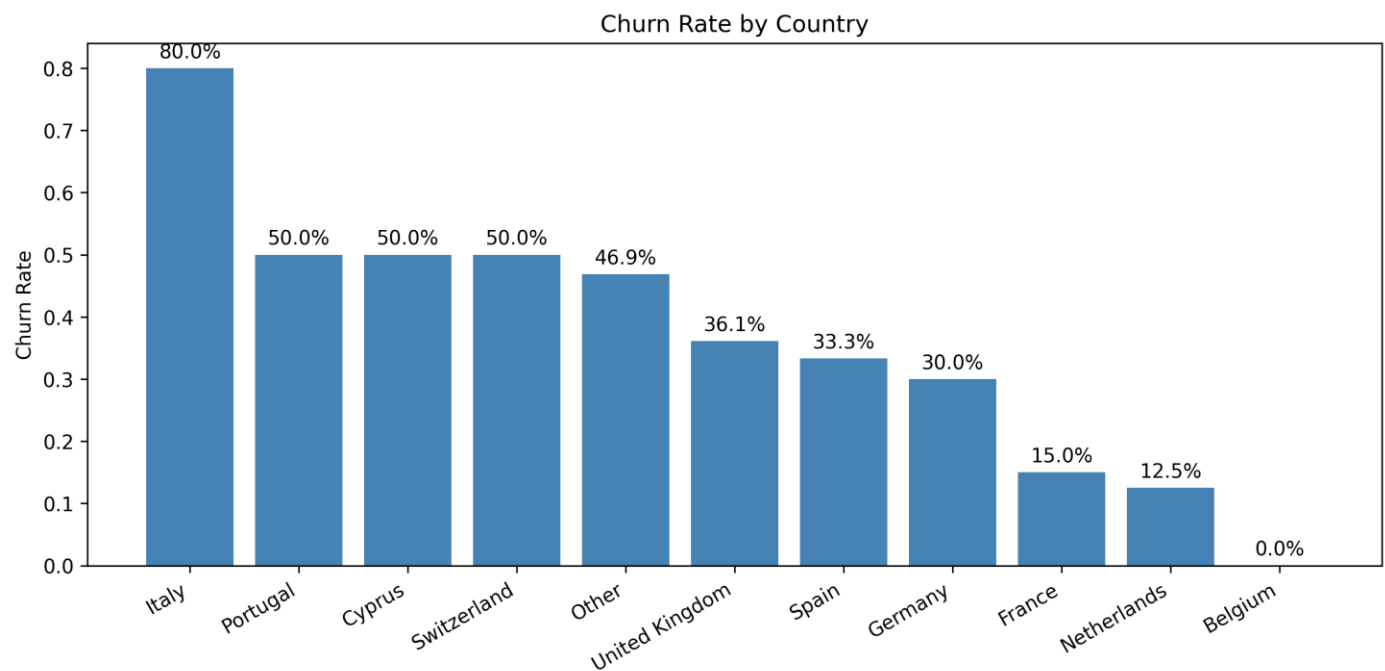


Figure 16 — Churn Rate by Country

The United Kingdom accounts for 1,574 of 1,692 customers (93%). All other countries have fewer than 32 customers — churn rates for individual non-UK markets are statistically unreliable and should not drive country-level decisions.

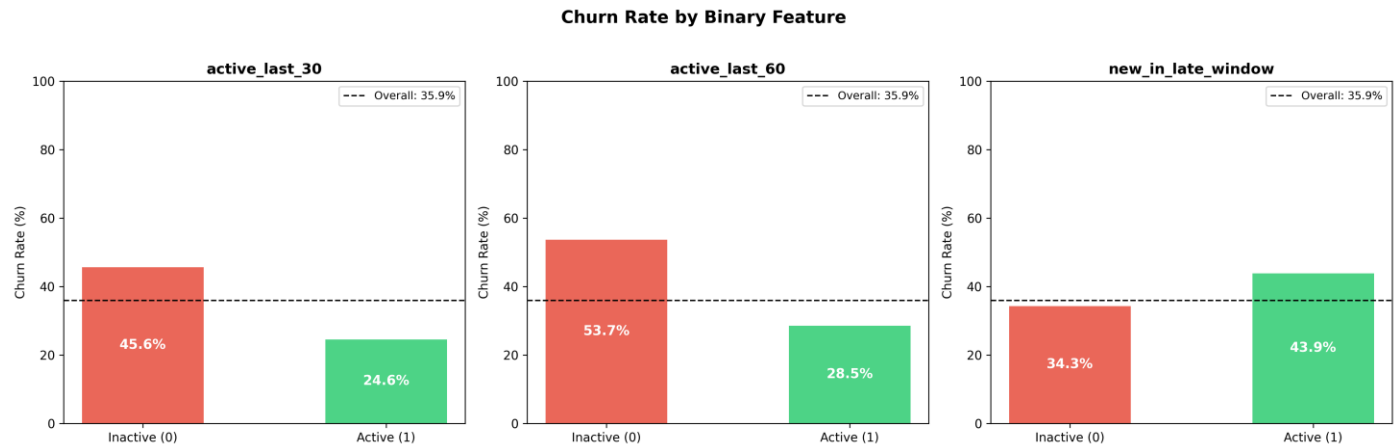


Figure 17 — Churn Rate by Activity Flag

The 60-day rule — the single strongest finding in this entire project:

Segment	Churn Rate	Lift vs baseline
Overall baseline	35.9%	1.00x
Active in last 60 days	28.5%	0.79x
Inactive for 60 days	53.7%	1.49x
Active in last 30 days	24.6%	0.68x
Inactive for 30 days	45.6%	1.27x
New in late window	43.9%	1.22x

A customer who has not purchased in 60 days will churn more than half the time. This rule requires no model to act on.

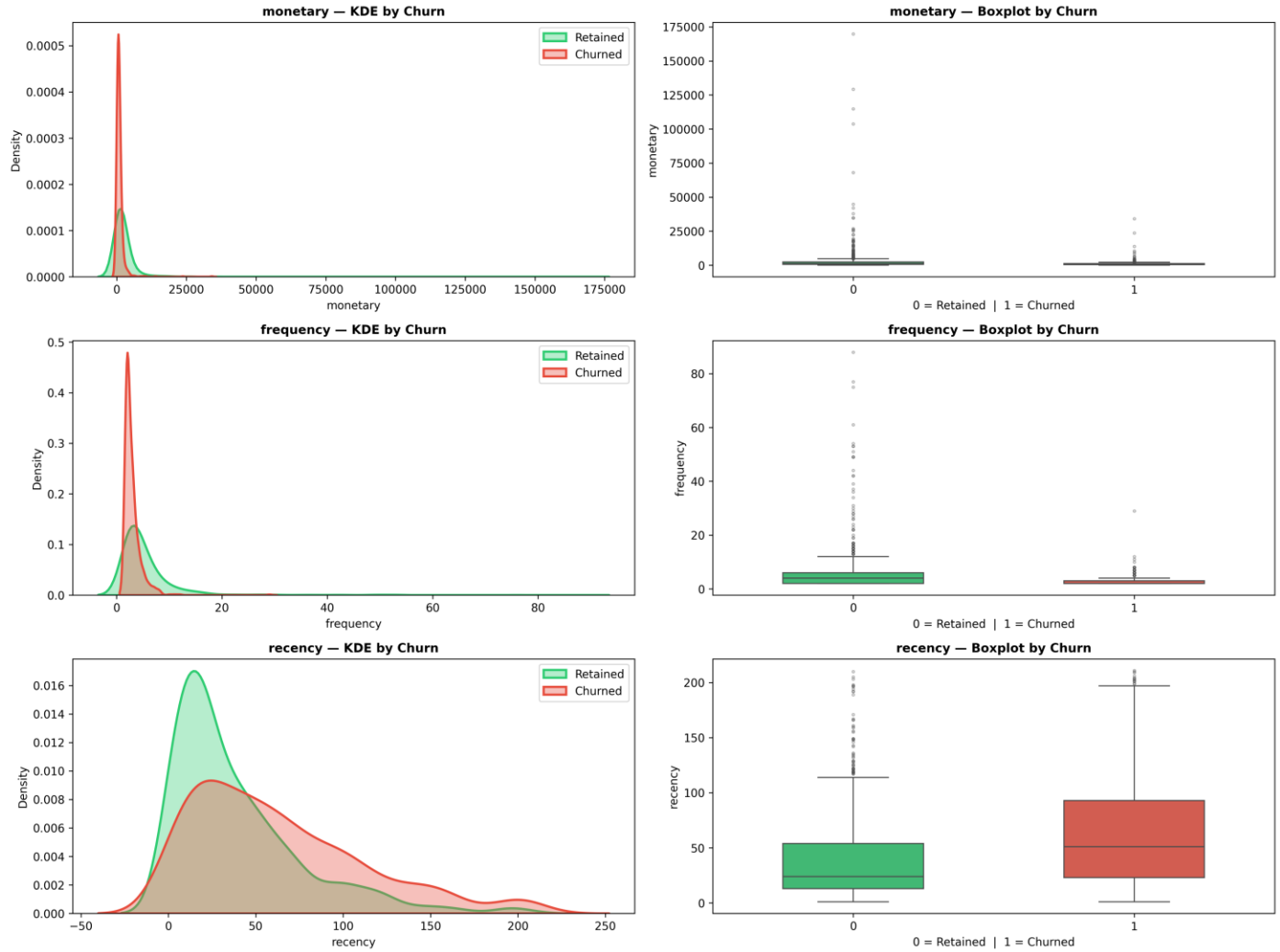


Figure 18 — Bivariate KDE and Boxplot: Monetary, Frequency, Recency vs Churn

Feature comparison between churned and retained customers:

Feature	Retained	Churned	Difference
Monetary spend	£2,900	£1,044	-64.0%
Frequency (orders)	5.75	2.93	-49.1%
Recency (days since last order)	39 days	63 days	+61.4%
Months active	3.0	2.0	-27.0%
Spend trend	1.23	0.88	-28.7%
Return rate	0.240	0.220	-8.5%

Churned customers spent 64% less, ordered 49% less often, and last purchased 61% more recently than retained customers. The spend trend finding is particularly important — customers who were already spending less in the second half of the observation window (spend_trend < 1) were systematically more likely to churn. Churn is a gradual decay, not a sudden event.

Return rate shows almost no difference between groups (-8.5%). **High return rates do not predict churn.**

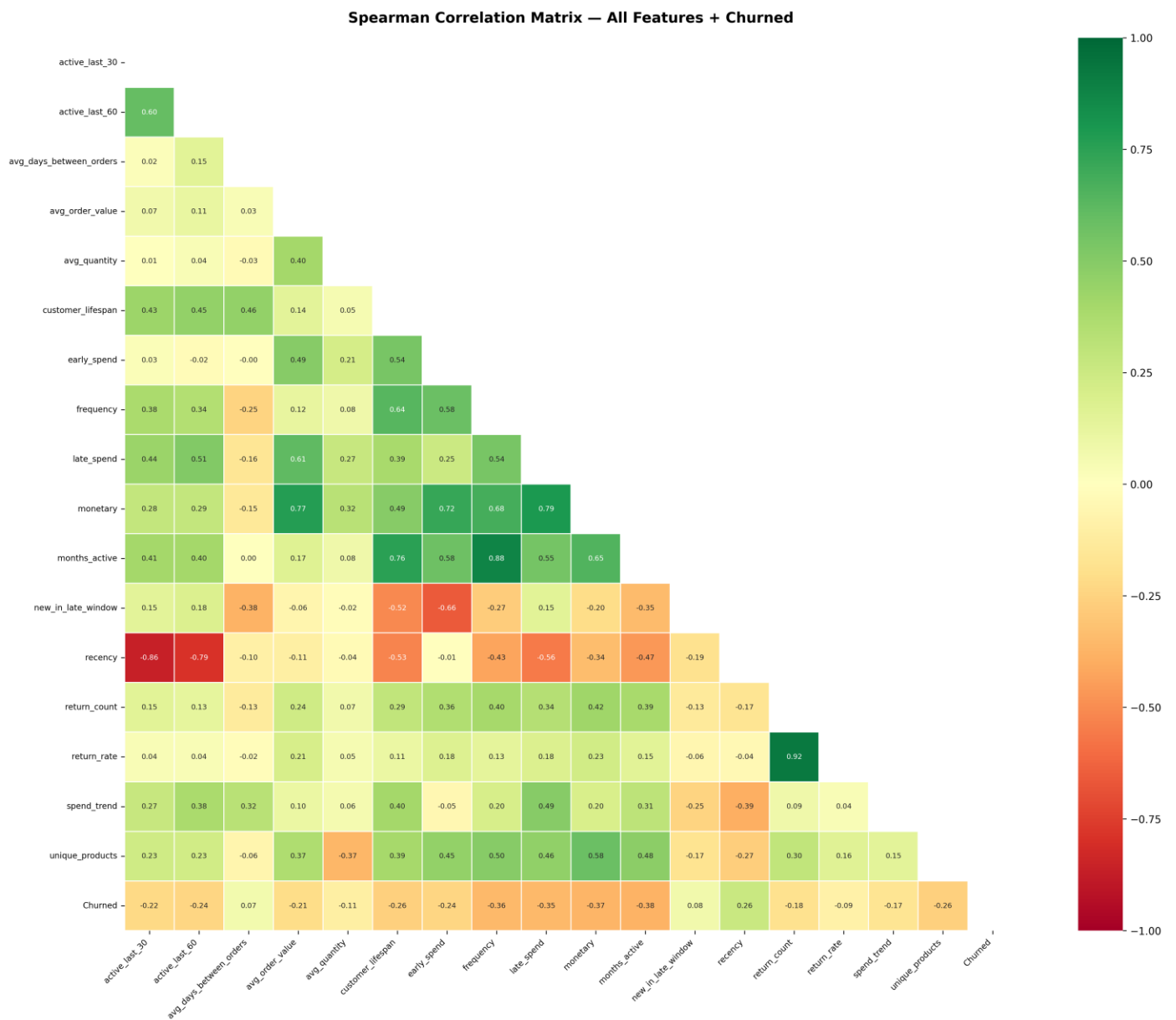


Figure 19 — Spearman Correlation Heatmap

7.3 Model Results

Three algorithms were evaluated in sequence. All three converged at approximately 0.77 ROC-AUC regardless of complexity — indicating the data ceiling has been reached, not a modelling failure.

Model	ROC-AUC	Precision	Recall	Revenue Capture	Pareto Capture
Naive baseline (60-day rule)	—	53.71%	—	—	—
Logistic Regression	0.7736	52.54%	76.23%	54.27%	25.00%
Random Forest (tuned)	0.7686	51.23%	75.41%	47.95%	0.00%
XGBoost (tuned)	0.7772	49.18%	75.41%	47.98%	0.00%
High-Value LR (dedicated)*	0.7439	16.67%	33.33%	71.44%	100%

**Evaluated on a separate high-value test set of 68 customers — not directly comparable to the general models above. Precision/recall shown at the default 0.50 threshold; the Pareto capture and revenue capture figures reflect performance at the recommended 0.25 threshold (see Section 7.5).*

Winner on business metrics: Logistic Regression (general model)

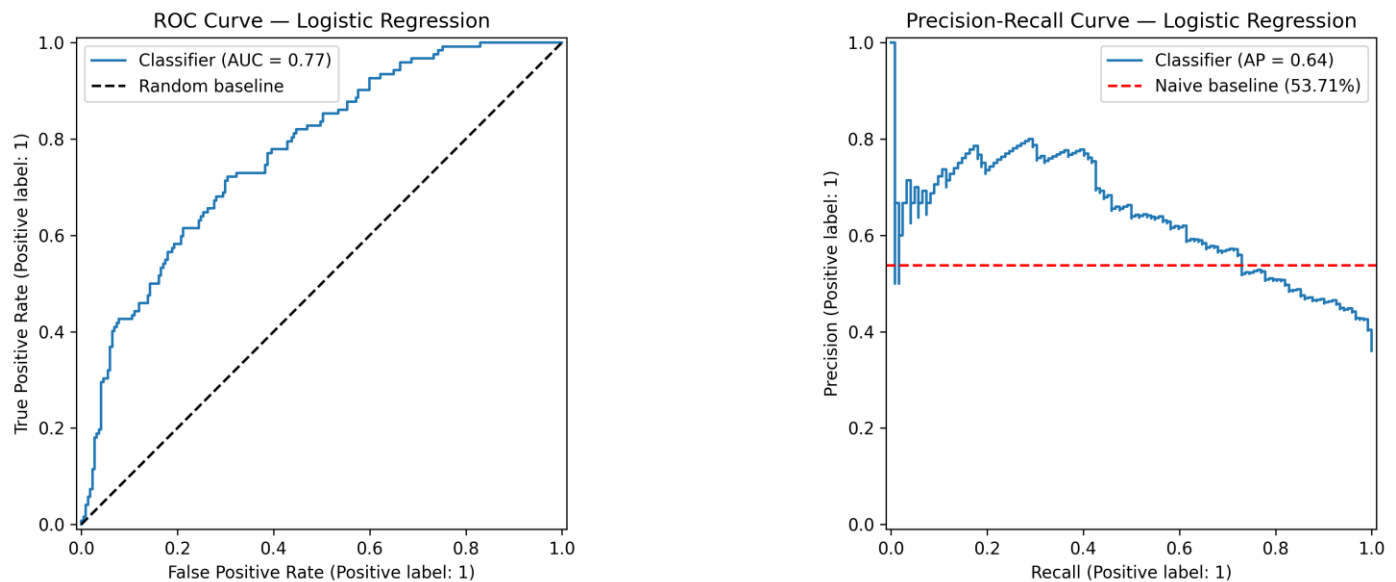


Figure 20 — ROC and Precision-Recall Curves — Logistic Regression

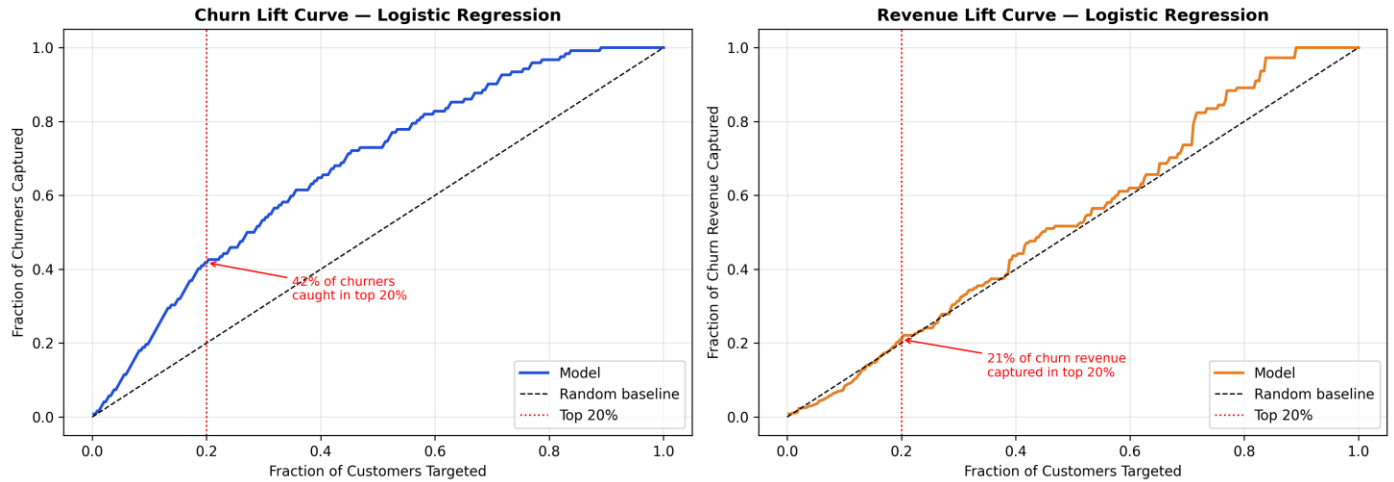


Figure 21 — Churn Lift and Revenue Lift Curves — Logistic Regression

Why tree models underperformed: Random Forest required `min_samples_leaf=40` to prevent overfitting on 1,353 training rows. This aggressive regularisation forced the model to require 40 similar customers per leaf — high-value customers, being rare by definition, never formed groups of 40, so the model lumped them with regular customers and predicted retention. Result: 0% Pareto capture. XGBoost showed the same pattern. Logistic regression, despite being the simplest algorithm, outperformed both on the metrics that actually matter to the business.

7.4 Dedicated High-Value Model

The top 20% of customers by revenue (spending above £2,199) represent 67.4% of total revenue. When the general model was evaluated on this segment it captured only 25% of high-value churners. A dedicated model was trained exclusively on these 339 customers.

Metric	General LR	High-Value LR
Revenue capture	54.27%	71.44%
Revenue missed	£49,548	£20,995
Improvement	—	+£28,553 less missed

The 17-percentage-point improvement in revenue capture comes purely from segmentation — same algorithm, same features, trained on the right population.

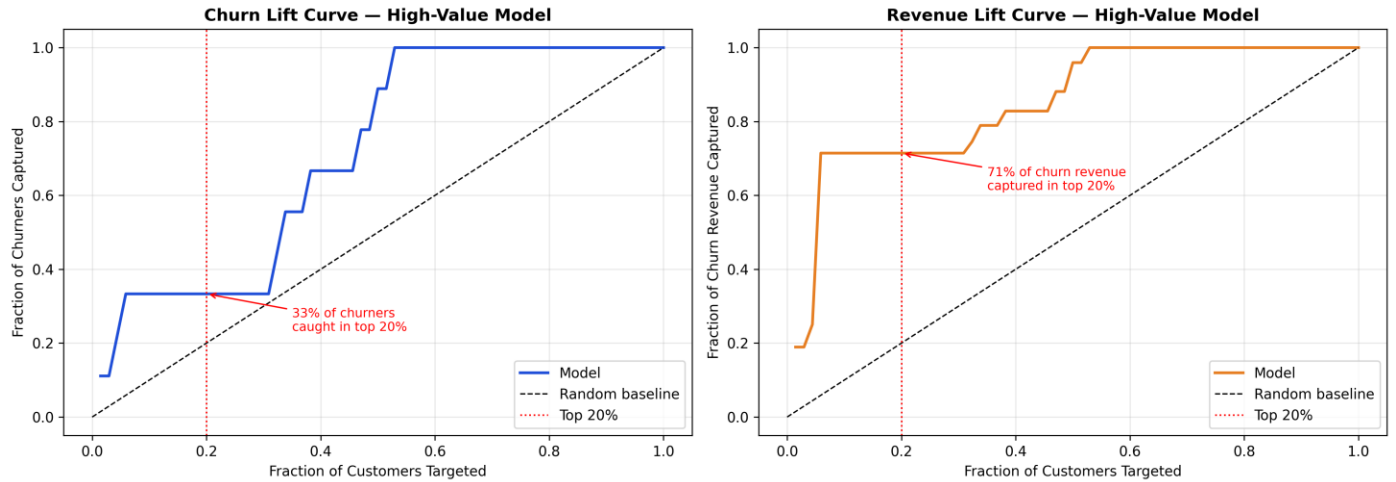


Figure 22 — Churn Lift and Revenue Lift Curves — High-Value Model

7.5 Threshold Optimisation

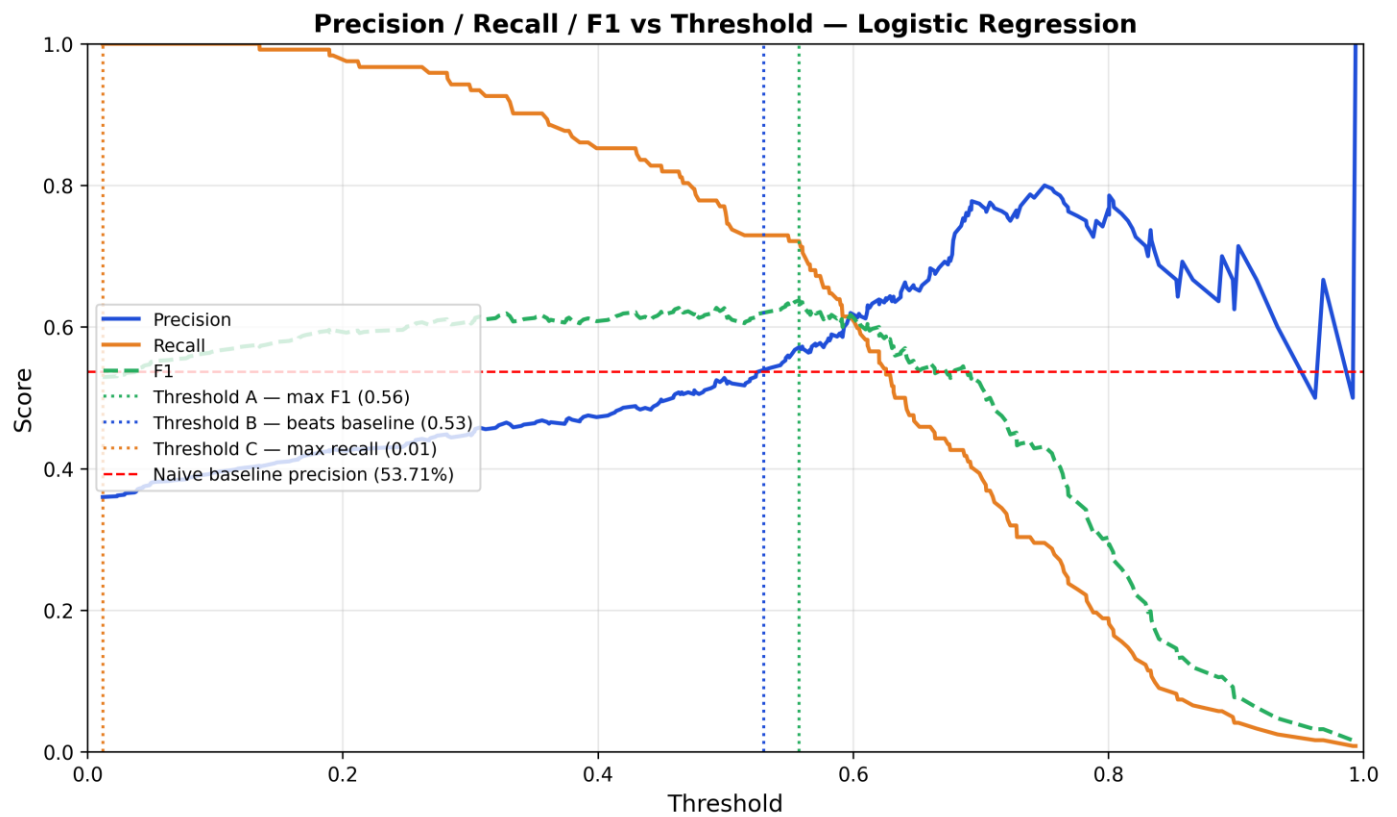


Figure 23 — Precision / Recall / F1 vs Threshold — Logistic Regression

Recommended thresholds:

Model	Threshold	Precision	Recall	Rationale
General LR	0.50	52.54%	76.23%	Highest ERS at £20,581
High-Value LR	0.25	~17%	100%	Never miss a high-value churning

The high-value threshold should never be raised above 0.25. The cost of contacting a false alarm in this segment (£10) is trivial compared to the revenue cost of missing a customer worth £10,000+.

Section 8 — Customer Lifetime Value

8.1 CLV Target Distribution

CLV was defined as actual customer spend in the prediction window (July — December 2010).

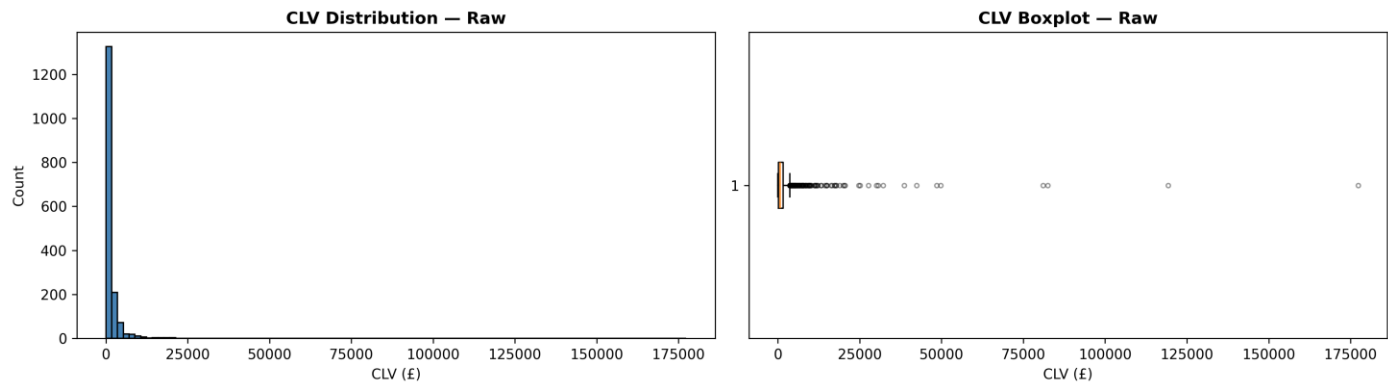


Figure 24 — CLV Raw Distribution: Histogram and Boxplot

Statistic	Value
Total future revenue captured	£3,089,132
Mean CLV	£1,825
Median CLV	£684
Std deviation	£6,798
Skewness	16.1
Customers with zero CLV	307 (18.1%)
Maximum single customer CLV	£177,334

The mean (£1,825) is 2.7x the median (£684) because a small number of whale customers dominate the aggregate. Standard deviation of £6,798 is nearly 4x the mean — the spread of values is enormous.

8.2 Revenue Concentration

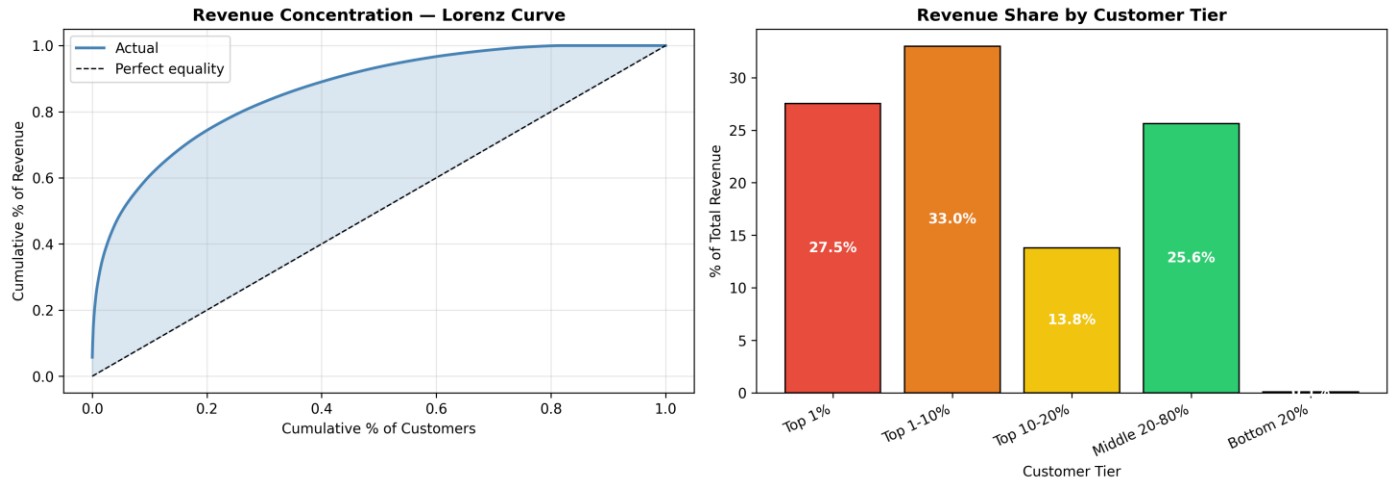


Figure 25 — Revenue Concentration: Lorenz Curve and Tier Bar

Customer Group	People	Revenue Share
Top 1%	16 customers	28.2%
Top 5%	84 customers	49.3%
Top 10%	169 customers	60.6%
Top 20%	338 customers	74.4%
Bottom 50%	846 customers	6.4%

16 customers generate 28% of all future revenue. 84 customers control half. This is more extreme than the classic 80/20 Pareto rule. The Lorenz curve hugs the bottom-right corner before shooting up steeply — the visual signature of extreme revenue concentration.

8.3 Feature Signal for CLV

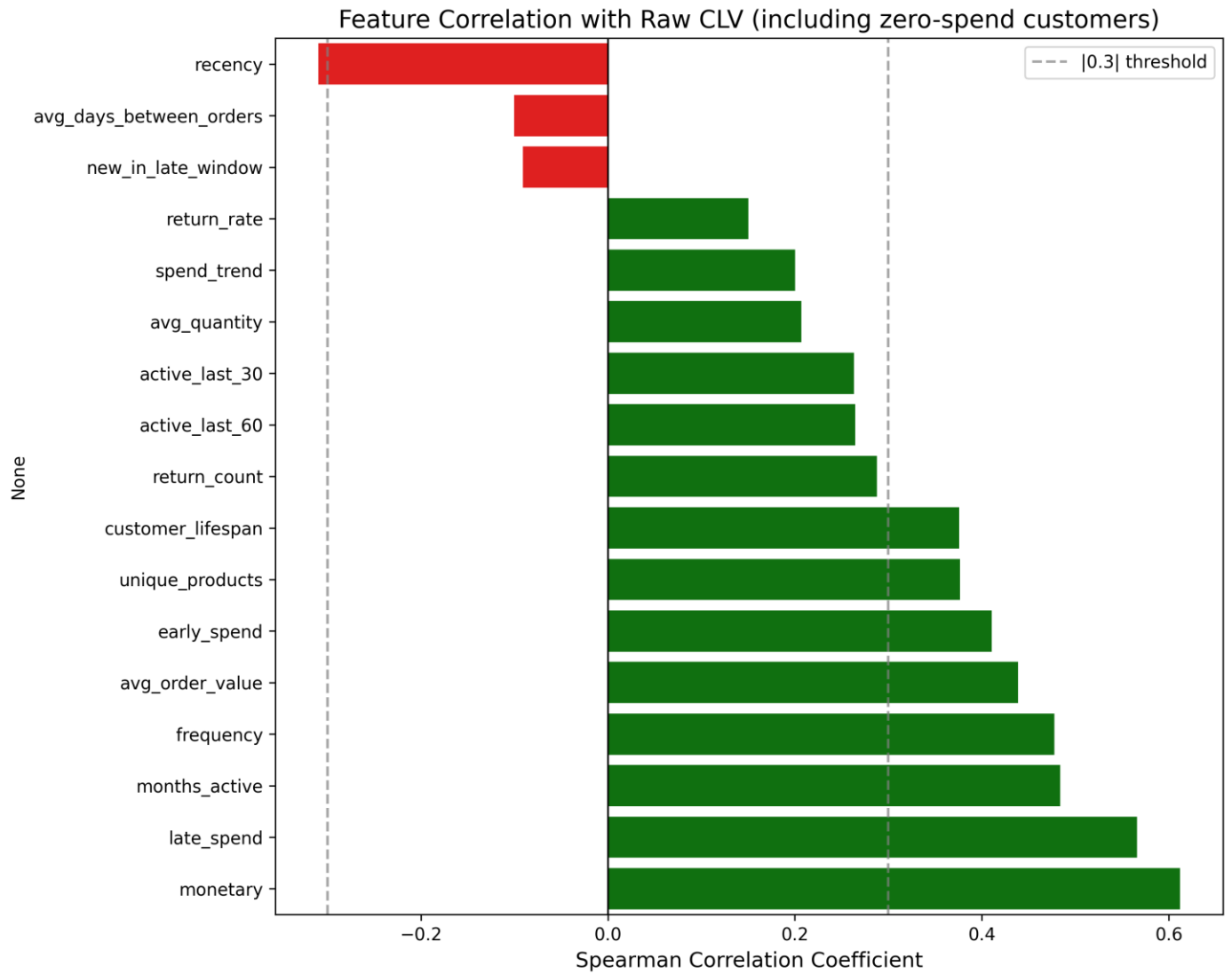


Figure 26 — Feature Correlation with CLV

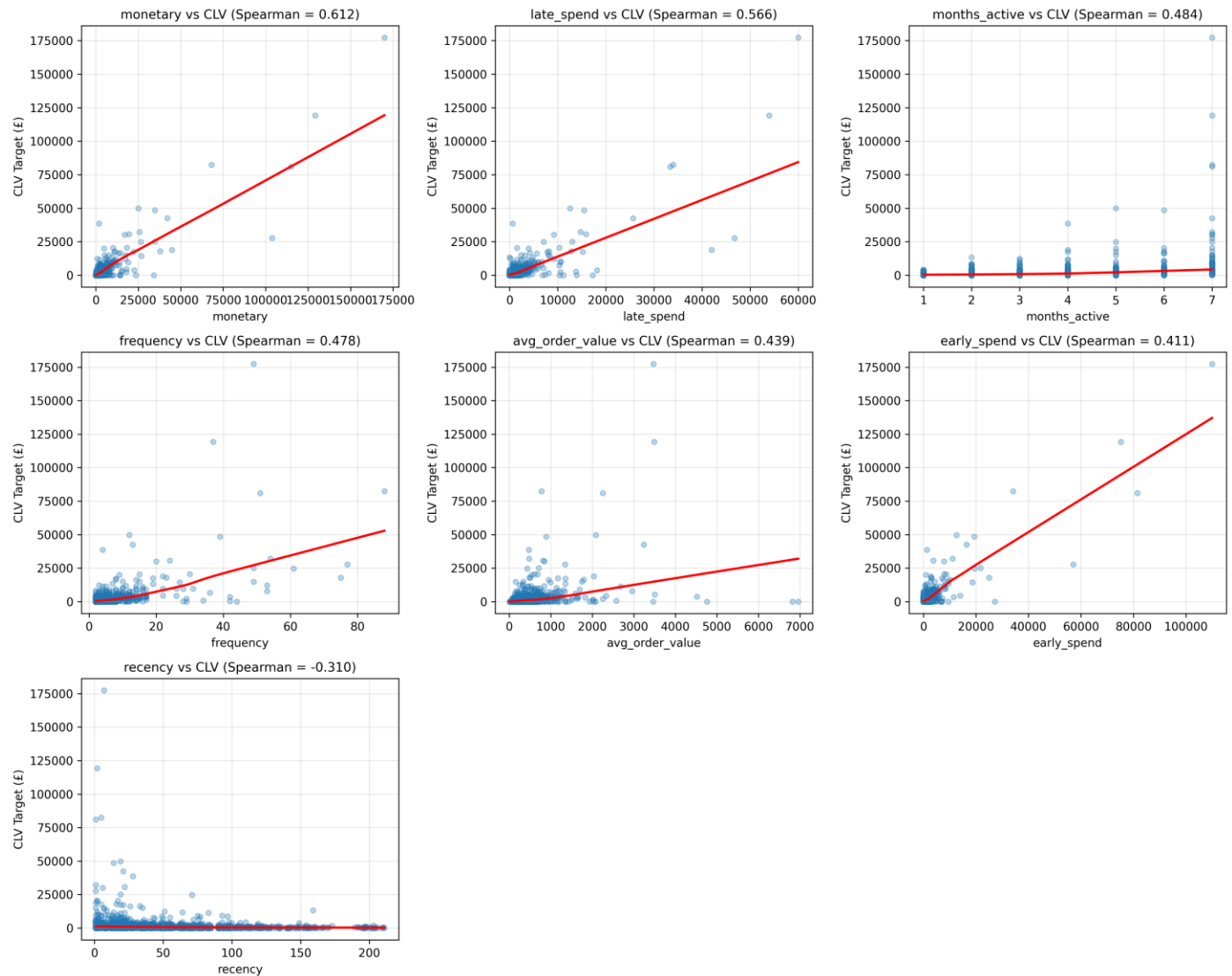


Figure 27 — Scatter Plots: Top Features vs CLV (Monetary, Frequency, Recency, Spend Trend)

Spearman correlation was used (not Pearson) because both features and target are heavily right-skewed. Pearson is distorted by the £177,334 customer — Spearman converts to ranks first, making it robust.

Feature	Spearman Correlation	Interpretation
monetary	+0.612	Strongest predictor
late_spend	+0.566	Second strongest
months_active	+0.484	Consistent engagement predicts future spend
frequency	+0.478	Purchase habit persists
avg_order_value	+0.439	Higher basket size continues

Feature	Spearman Correlation	Interpretation
recency	-0.310	More recent = higher future spend
avg_days_between_orders	-0.100	Weak signal — near zero

early_spend and late_spend were dropped from the feature set due to near-perfect correlation with monetary (0.969 and 0.940 respectively). Keeping all three would split model importance without adding signal and destabilise linear models.

8.4 Model Results

Model	MAE	RMSE	R ²	Revenue Capture	Pareto Capture	Tier 2 Capture
Mean predictor (floor)	£2,197	£7,698	—	—	—	—
Two-stage (Churn gate + LinReg)	£1,599	—	-0.832	73.7%	65.1%	—
Random Forest	£1,422	£4,606	0.641	82.5%	86.0%	9.6%
XGBoost default	£1,668	£6,734	0.233	83.1%	90.7%	—
XGBoost tuned	£1,464	£4,637	0.636	84.4%	93.0%	9.0%
CatBoost (final)	£1,429	£4,320	0.684	84.6%	90.7%	15.5%
CatBoost tuned	£1,378	£4,342	0.681	85.1%	90.7%	—
Two-model CatBoost	£1,426	£4,322	0.684	83.3%	86.0%	15.1%

Winner: CatBoost (untuned). Best R² and RMSE of any model. Lowest wasted spend. Tier 2 capture nearly double any other model at 15.5%. Tuning actually degraded R² from 0.684 to 0.681 because GridSearch optimised for MAE — it traded variance explanation for point accuracy.

Why CatBoost beat XGBoost on this dataset: Ordered boosting computes gradients on held-out portions of training data, reducing variance naturally. On datasets under ~5,000 rows this matters significantly. Always try CatBoost first on small tabular problems.

Why the two-stage architecture failed: The churn model was trained on the assessment-window inactivity definition described in Section 7.1. CLV zeros are defined by a 6-month purchase window. The model thinks 35.9% churned; reality is only 18.1% have zero spend. No threshold can fix a model trained on the wrong definition of zero.

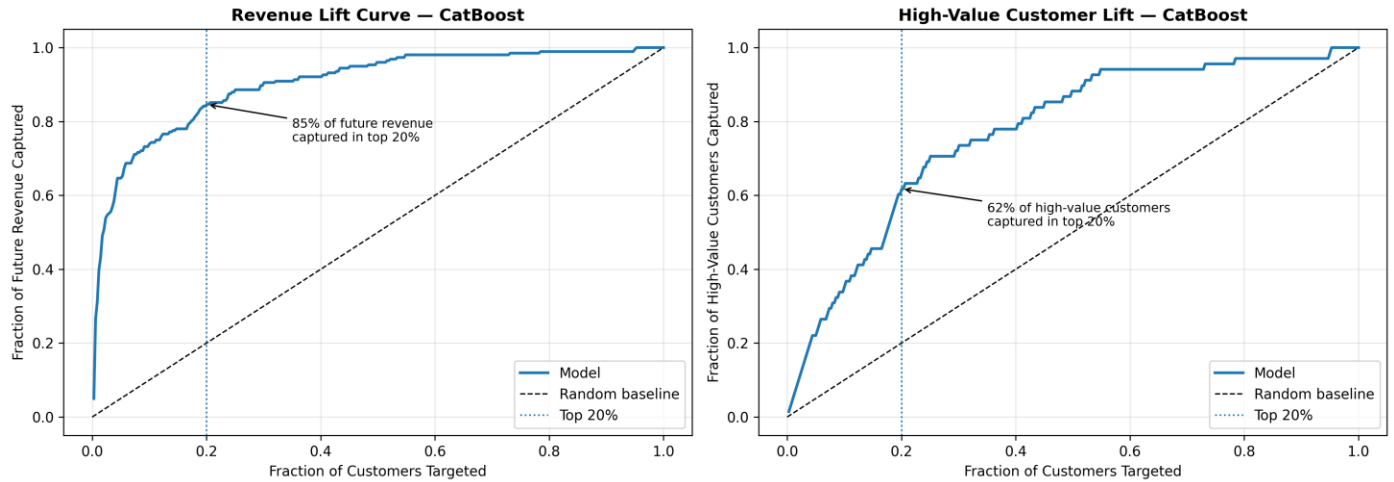


Figure 28 — CatBoost Revenue Lift and High-Value Customer Lift

8.5 Why Tier 2 Capture Is Stuck at 15%

Every architecture tested — Random Forest, tuned XGBoost, CatBoost, two-model — lands Tier 2 capture between 9% and 15%. This convergence is the clearest signal that a data ceiling has been reached.

Mid-value customers (roughly £200–£2,000 future spend) look identical in transaction feature space. A customer who spent £600 and one who will spend £1,800 have overlapping frequency, recency, monetary, and spend trend profiles.

What would fix this: a customer type flag (B2B vs individual), product category diversity scores, firmographic data for business accounts, or acquisition channel information. The two-model architecture is already built — it just needs features that actually separate mid-value customers.

Section 9 — Integrated Action Matrix

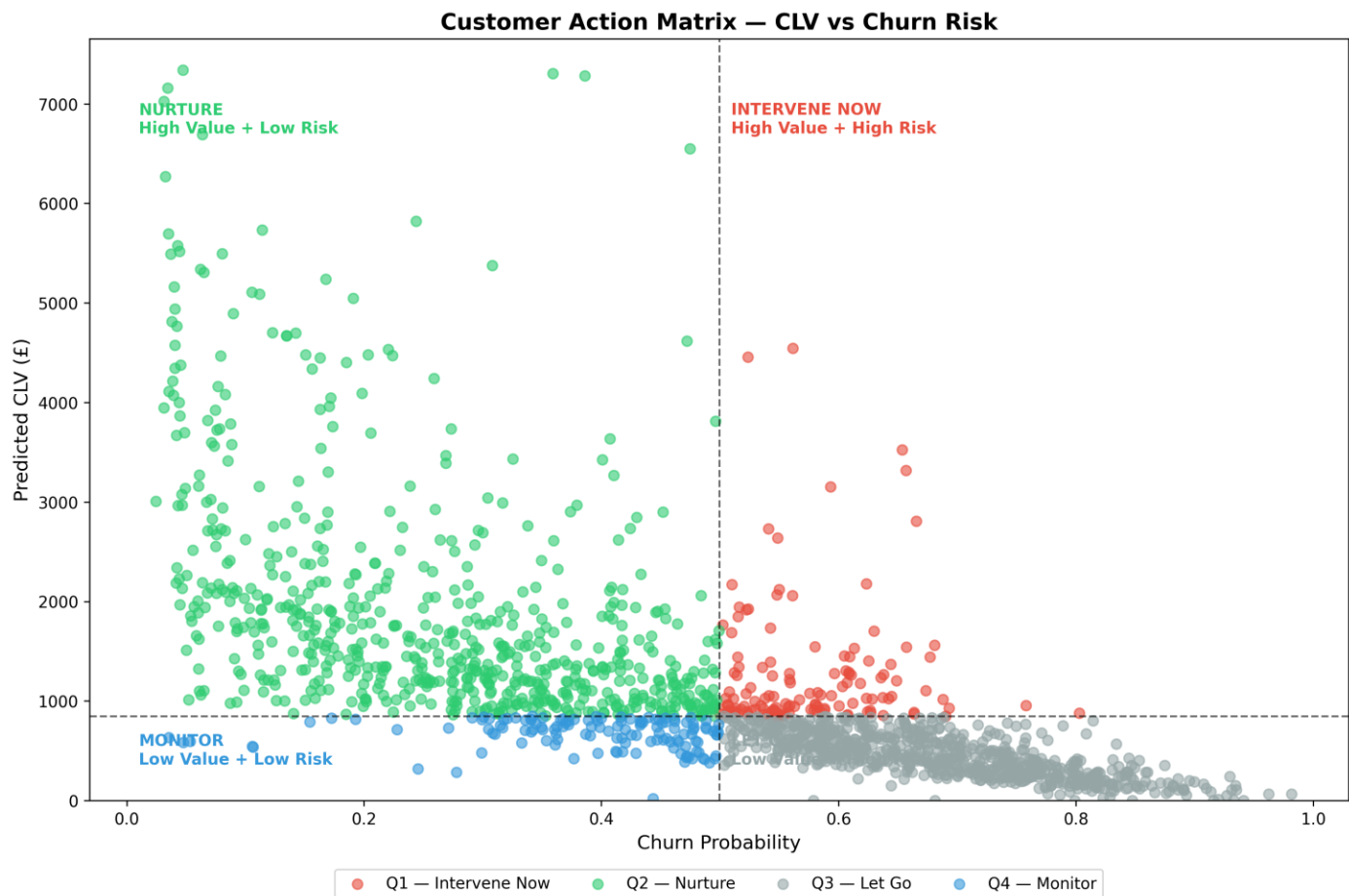


Figure 29 — Customer Action Matrix: CLV vs Churn Risk

The action matrix combines CatBoost CLV predictions with churn model probabilities across all 1,692 customers. Quadrant boundaries are set at the medians of predicted CLV (£848) and churn probability (0.50).

Quadrant	Customers	Total CLV	Avg CLV	Avg Churn Risk	Priority
Q1 — Intervene Now	132	£178,323	£1,351	High	Immediate
Q2 — Nurture	714	£2,496,022	£3,496	Low	Ongoing
Q3 — Let Go	714	£323,945	£454	High	Minimal
Q4 — Monitor	132	£88,757	£672	Low	Passive

Reading note for the client: Q2 customers show a higher average CLV than Q1. This is correct and expected — high-CLV customers with low churn risk are by definition your most valuable stable accounts. Q1 customers are urgent not because they are the most valuable overall, but because they are valuable and already leaving. The urgency is time, not absolute value.

Section 10 — Business Recommendations

Priority 1 — Deploy the 60-Day Reactivation Campaign (Act this week)

Customers inactive for 60 days churn at 53.7%. This requires no model to implement. Set the trigger at day 45 — before the relationship breaks, not after.

- Day 45: Personalised re-engagement email referencing their last purchase specifically. Time-limited incentive to return.
- Day 60: Escalate. Phone call for high-value customers. Stronger incentive for regular customers.

Even a 20% success rate on reactivated customers represents material revenue recovery per period.

Priority 2 — Fix the Onboarding Process (Act this month)

New customers churn at 43.9% — nearly double the baseline of 35.9%. Nearly half of all newly acquired customers are leaving before becoming established. Marketing spend acquiring them is being partially wasted.

- Day 7: Welcome email highlighting top products in their purchase category.
- Day 30: Check-in with first-time buyer discount on second purchase.
- Day 60: If no second purchase, treat as high churn risk and trigger a win-back offer.

Priority 3 — Launch a VIP Retention Programme (Act this month)

The top 20% of customers by revenue account for 67.4% of total revenue (£2,547,864). The dedicated high-value model identifies these customers with a CV ROC-AUC of 0.83. At threshold 0.25, revenue capture reaches 71.44%.

Monthly process: identify all customers spending above £2,199 → run through the high-value model → flag anyone with churn probability above 25% for personal outreach.

Financial case:

- Intervention cost: £360 per period (36 contacts × £10)
- Revenue identified at risk: £73,506
- Expected revenue saved at 35% retention: £25,727
- Net return: £25,367 — approximately £71 returned per £1 spent

Priority 4 — Deploy the Monthly Prediction System (Next month)

Run both models monthly on updated customer data:

4. High-value model at threshold 0.25 → VIP retention (personal outreach)
5. General LR model at threshold 0.50 → Standard retention (email campaign)

Combined expected return per period at 35% retention:

- General model ERS: £20,581
- High-value model ERS: £25,727

- Total intervention cost: £2,140
- Net return: £44,168

Priority 5 — Rethink the Returns Policy (This quarter)

Retained customers have significantly higher return counts than churned customers. Returns are a loyalty signal, not a cost signal. Customers who return items are engaged and trust the process enough to use it.

Do not penalise customers for returning items. Do not restrict return allowances for frequent returners. If the business currently suppresses communications or reduces incentives for high-return customers, it may be actively churning its most engaged buyers.

Priority 6 — Invest in Data Collection (Ongoing)

All three models converged at approximately 0.77 ROC-AUC regardless of algorithm complexity. This ceiling is the data, not the modelling. Two specific investments would break through it:

Confirmed churn labels: Add a cancellation event, account deletion, or explicit opt-out to the platform. Six months of confirmed churn labels are expected to push ROC-AUC from 0.77 toward 0.83–0.87.

High-value churner volume: The dedicated model trained on 34 churners — statistically fragile. Collect until 200+ high-value churn events are recorded. Given £2,547,864 in high-value revenue at stake, this data collection has an extremely high ROI.

Priority	Action	Requires Model?	Timeline
1	60-day reactivation campaign	No	This week
2	Fix onboarding sequence	No	This month
3	VIP retention programme	Yes — HV model	This month
4	Monthly prediction system	Yes — both models	Next month
5	Review returns policy	No	This quarter
6	Data collection initiative	No	Ongoing

Section 11 — Technical Appendix

A. Churn Feature Engineering

Observation window: 1 December 2009 — 30 June 2010 **Churn assessment window:** 1 July 2010 — 30 September 2010 **Features engineered:** 28 (reduced to 16 after feature selection)

Core features retained: recency, frequency, monetary, unique_products, avg_quantity, customer_lifespan, avg_order_value, early_spend, late_spend, return_count, months_active, active_last_30, active_last_60, new_in_late_window, spend_trend, return_rate

Preprocessing: Yeo-Johnson transformation applied to 10 skewed features (all brought within ± 0.3 of zero skew). StandardScaler applied for linear models. Country one-hot encoding dropped after confirming statistical unreliability outside UK.

B. CLV Feature Engineering

Observation window: 1 December 2009 — 30 June 2010 **Prediction window:** 1 July 2010 — 9 December 2010 **Features:** 15 (early_spend and late_spend dropped due to 0.97 and 0.94 correlation with monetary)

Train/test split: 80/20 stratified on CLV tier (Zero/Low/Mid/High) to ensure all tiers represented proportionally.

C. Revenue Forecasting Methodology

Models tested: Naive (last observation), Moving Average (windows 2, 3, 4), Holt exponential smoothing with damped trend

Walk-forward validation: Starting at month 7, each model trained on all preceding months and tested on the next single month. Six validation points total.

Outlier handling: November 2010 excluded from conservative scenario ($z=2.60$). December 2010 excluded from all scenarios (confirmed data artefact — partial month).

Confidence interval method: Empirical percentiles of 6 signed forecast errors. With 6 errors the 2.5th and 97.5th percentiles are the minimum and maximum errors respectively. This is not a statistically robust interval — it is an observed error range. A reliable CI requires 24+ months.

D. Model Files Produced

File	Contents
churn_model_general_production.pkl	LR churn model — full 1,692 customer training
churn_model_highvalue_production.pkl	LR high-value churn model — 339 customers
churn_predictions_december_2011.csv	Production predictions with risk tier and recommended action
clv_predictions.csv	Predicted CLV + CLV tier for all 1,692 customers

File	Contents
action_matrix.csv	CustomerID + predicted CLV + churn probability + segment label